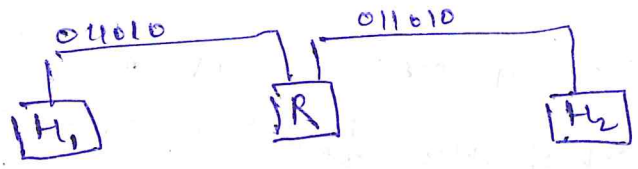
Network devices:

Repeater

→ Physical-layer device.

→ 2 port.



Copies data to original-signal-strength.

→ Repeaters are used to re-gain the strength of the signal.

→ If the signal is weak and is not reaching to other host. We use repeater to place it in-between two hosts.

→ It only works on encoding of data and repeats the signal.

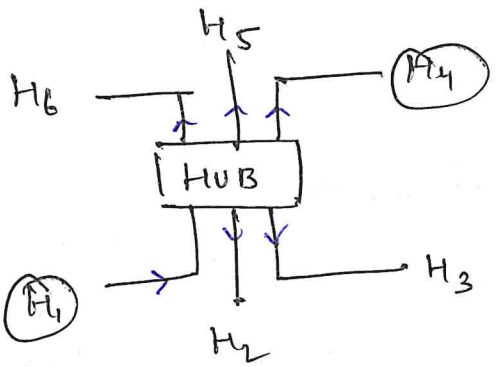
HUB :- → physical layer device

→ Multiport - repeater.

→ no-intelligence.

→ Cannot filter data

→ Cannot find best paths.



It transfer the data in all directions, not an intelligent device.

* It just copies the data from one port and repeat the data to all other ports by maintaining the original strength of signal.

Active : Own power supply. Repeater + wiring - center.

Passive : Cannot repeat, just relay signals.

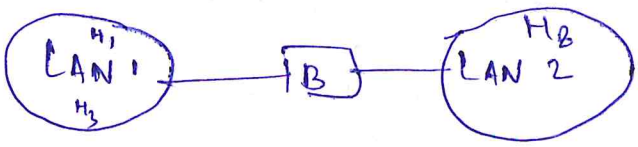
Bridge :

→ DLL - device → frame, MAC → Ethernet, Token ring, WLAN

→ Connect 2 - LANS

→ filter based on MAC

→ Transparent vs Source - routing bridges



* It uses mac address to filter. If H_1 send data to H_B. Bridge forward to LAN 2.

Source routing bridges exactly follows the path $H_1 - H_6 - \ldots - B \rightarrow H_7 - H_8$ suggested by the source.

Switch :

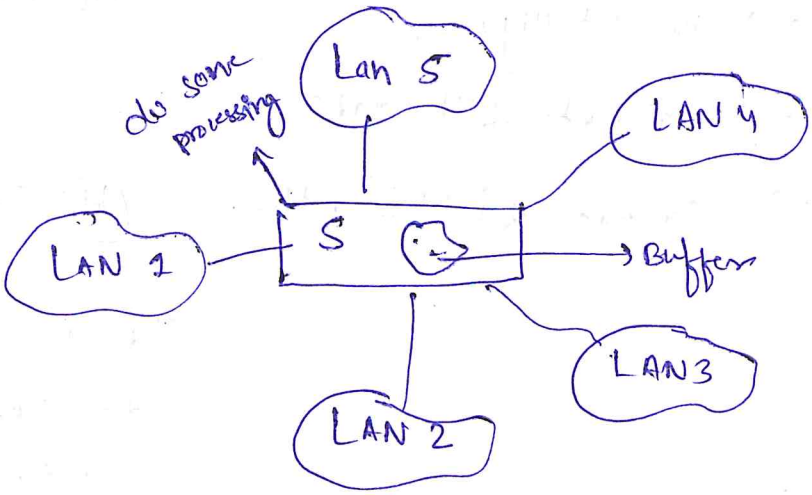
→ Multiport-bridge

→ Buffers

→ DLL device

→ error check before forwarding

→ forwards to the correct port



* Switch also uses MAC address for forwarding. It has buffer and some processing unit which checks whether sending frame is correct or not.

Routers :—

→ routes packets based on IP-addresses.

→ Network layer device.

→ routing-tables.

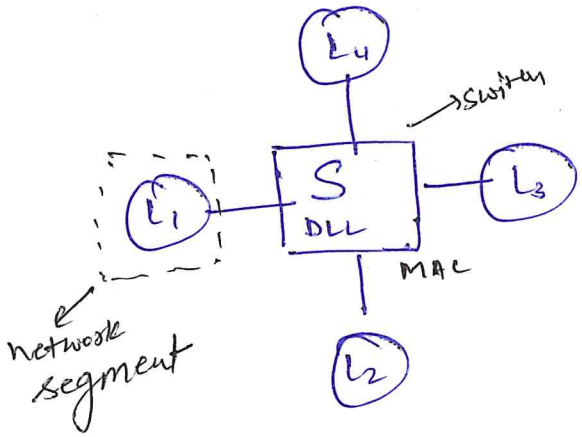
→ connect LANs and WANs.

Brouters:

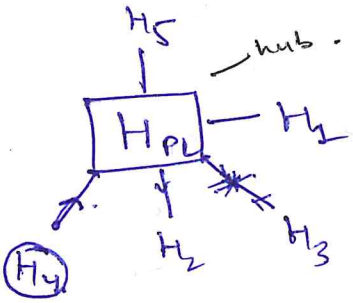
→ DLL and Network - Layer device.

* It performs task of both DLL and network layer.
* So, it can perform the task of both.

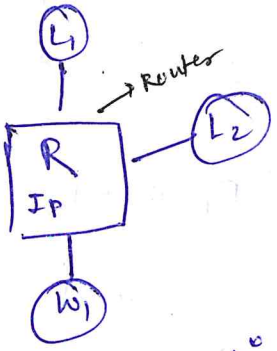
Most Popular devices: Switches and Routers



vs



vs



Network segment:- Network segment is collection of hosts

We can consider one LAN as one segment.

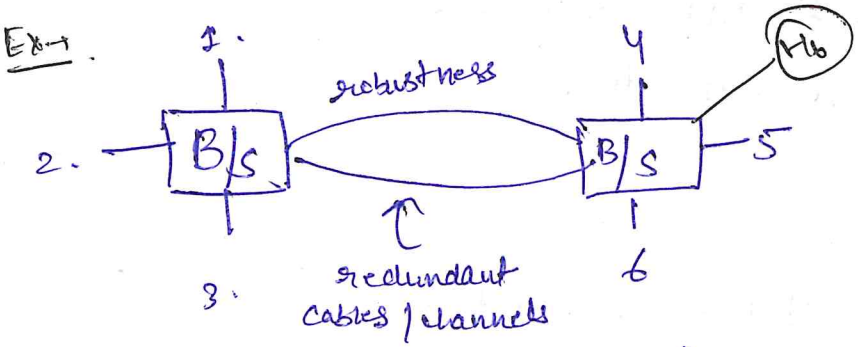
	Switch	Hub (not a smart device.)	Router.
Collision domain -	Seperator	Can not break	Seperator
Broadcast domain - (NW L IP)	Can not break.	Cannot break.	Seperator.

We want to broadcast in a LAN.

Spanning Tree Protocol (STP)

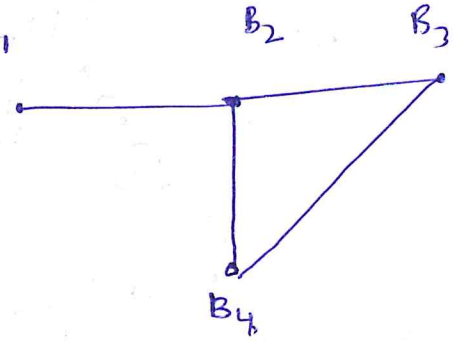
* We learn spanning tree in graph and Algorithms.

Ex-1.



→ we are using two cables for robustness / reliability.

Ex-1. 2



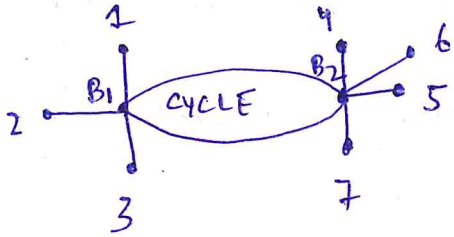
for $B_1 - B_4$ ⇒ $B_1 - B_2 - B_3 - B_4$ or $B_1 - B_2 - B_4$

* If one path goes bad still we are able to transmit via other path.

→ Unknown destination ⇒ flood ⇒ ∞ loops.

↳ If switch doesn't know about any host. It will flood the packet.

→ Spanning tree ⇒



→ You can encounter ∞ loops because of cycle.

* Spanning tree it covers all the vertices without loops.

Distributed ST-algo :—

→ root node : Smallest MAC [Smallest MAC address will be root node

→ Configuration frames

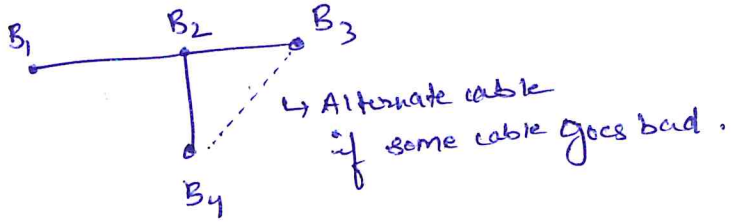
→ Shortest path from root

hops prop time.

* Shortest path from root can be determined on the basis of hops and prop time.

→ IEEE 802.1D → protocol will give you lots of details about how ST-algo format etc works.

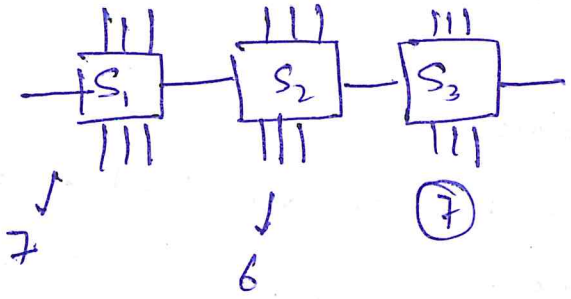
final spanning tree for Ex-2 is



→ Radia Perlman CSE Scientist developed this algorithm.

Que: Consider that 15 machines need to be connected in a LAN using 8-port Ethernet switches. Assume that these switches do not have any seperate uplink ports. The minimum number of switches needed is?

Sol: One switch has 8 connections.



Maximum we can connect = 20 using 3 switch.

For 15 we need minimum 3 switch.

Network layer :— (Internet protocol)

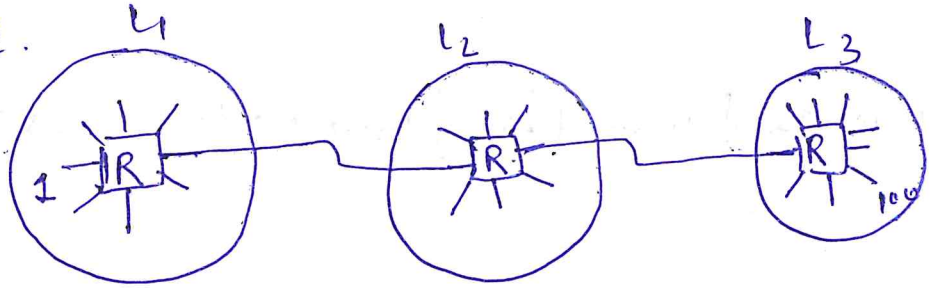
1. IP- address, Subnetting and Supernetting
2. IPV4 and IPV6
3. Routing algorithms.

IP Addressing (IPV4)

→ 32 bit logical address assigned to computer.

Q. Why not use MAC-addr?

Ans.



There are billion of hosts in the network. Each host has unique mac address.

There is no particular series of MAC address. Every company follows different rule.

If H_1 wants to send the packet to host 100. The router doesn't know any net-id or subnet-id. It is very difficult for router to communicate. So, if router floods the packet to every link this will increase lot of congestion.

We don't have hierarchical structure in MAC address.

Like postal address: State
↓
City
↓
Area → Street → house no.

Classfull Addressing

IP address = 32 bit. So, total = 2^{32} addresses possible

(A) | 0 | Network (7) | Host (24) | 1.0.0.0 to 127.255.255.255

(B) | 10 | Network (8+8) | Host (16) | 128.0.0.0 to 191.255.255.255

(C) | 110 | Network (5+16) | Host (8) | 192.0.0.0 to 223.255.255.255

(D) | 1110 | Multicast address | 224.0.0.0 to 239.255.255.255

(E) | 1111 | Reserved for future use. | 240.0.0.0 to 255.255.255.255

* In class A first bit is always zero, next 7 are net bits and 24 are host bits

* In Ip address we have four octets.

Range → 0-255 . 0-255 . 0-255 . 0-255
first octet | Second octet | Third octet | Fourth octet

→ In class A the range of first octet is
  * 1 - 126
  * 127 is is reserved as loop back address.

→ In class B 10 is the starting bits always.
In class B net bits (6+8) = 14 and host bits are 16

→ In class C 110 is the starting bits always
In class BC 21 bits as net bits and 8 bits are host bits.

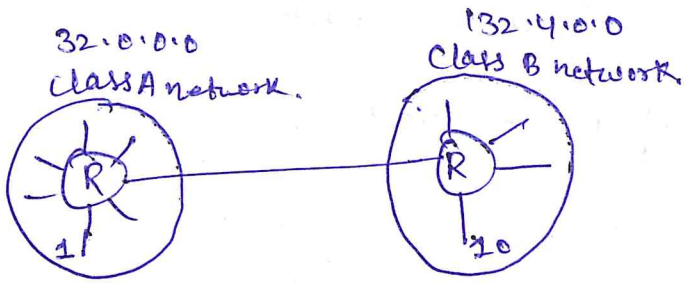
→ In class D 1110 is the starting bits. The class D addresses are used for multicasting.

→ In class E 1111 is the starting bits. The class E addresses are reserved for future use.

Ranges:

Class	First octet
A	1-127
B	128-191
C	192-223
D	224-239
E	240-255

UNICASTING :-



32.1.2.3

Class A address

132.4.5.6

Class B address

→ For getting the Ip address of the network, put all host bits at zero.

Host address → 32.1.2.3 [Class A address]

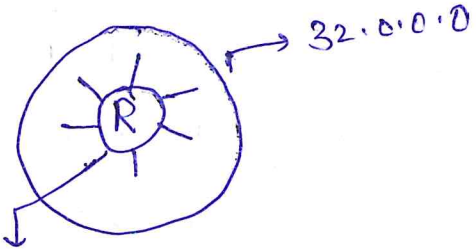
So, Network address → 32.0.0.0

host bits

→ In Unicasting we are transfering from single host to single host.

Broadcasting

Limited:



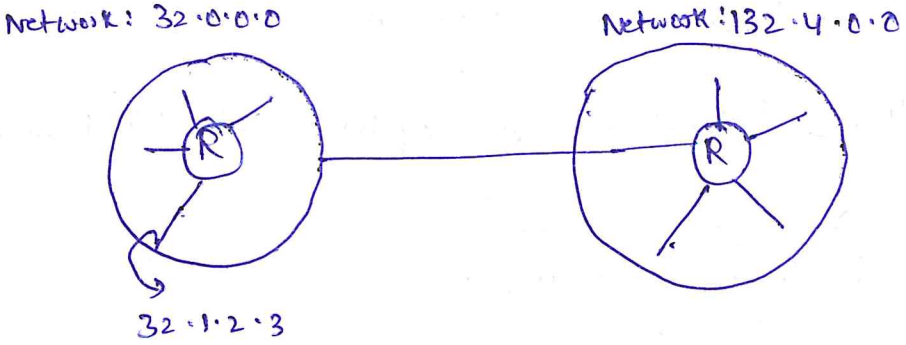
SA : 32.1.2.3

DA : 255.255.255.255

→ Limited broadcast is used to send the packet from single host to all the hosts within a network.

Here in destination address put all one's.

Directed broadcast :-



→ Directed broadcast is used to broadcast the packet in the different network.

→ We have to assign all host bit's as one's in the network address of destination network.

SA: → 32.1.2.3

DA: → 132.4.255.255

Host bits all one's.

Concept:- No. of hosts in a network.

Class A : 1.0.0.0 to 127.255.255.255 [Range]

* In class A eg: 32.0.0.0 [network address]

→ we have 24 host bits in class A

→ So, $2^{24} - (2)$ [Total number of host address possible]

** We are subtracting 2 because:

→ One address is waste in network address
32.0.0.0

→ Another address is waste in broadcast address
32.255.255.255

Class B 128.0.0.0 to 191.255.255.255 [Range]

e.g: 132.1.0.0 → $2^{16} - 2$.

→ we have 16 bits for host address

→ So, total number of address is $2^{16} - 2$ [host addr]

Class c : 192.0.0.0 to 223.255.255.255

e.g. 192.1.2.0 → $2^8 - 2 = 254$

8 host bits.

Subnetting :-

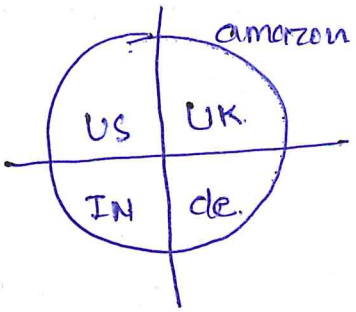
Wire-Shark : IP `52.94.232.33` — Class A [Amazon-server]

DNS

↓

New-IP: `52.0.0.0` ← for Net addr put host bits zero.

DBC IP : `52.255.255.255`



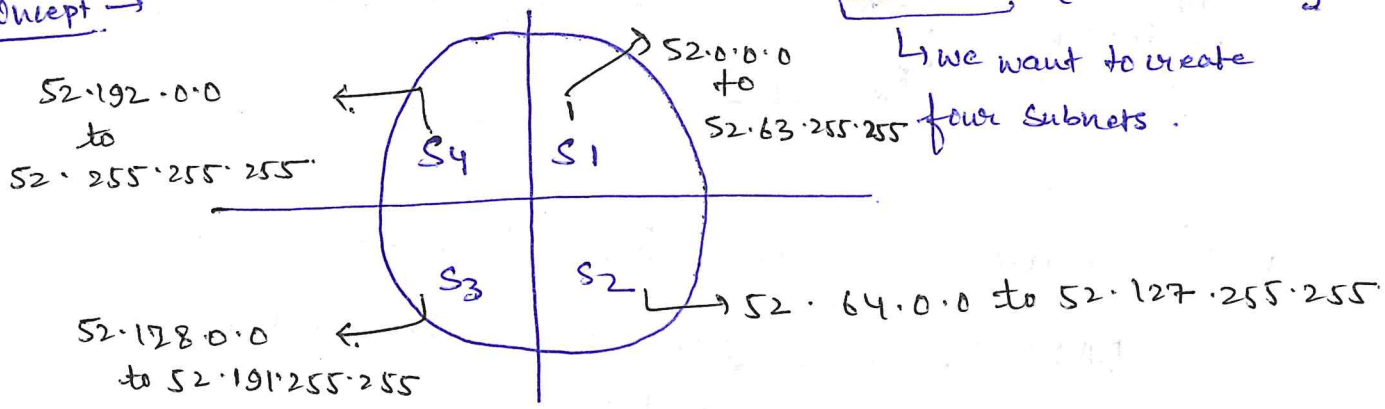
Max No. of hosts $= 2^{24} - 2 \approx 16$ Million.

* For legal and security reasons it's better to break a big network into sub-networks.

Concept →

`52.0.0.0` [Network addr]

↳ we want to create four subnets.



* In Subnetting we borrow some bits from hosts to create subnets.

`52.` ① ② ③ ④ ⑤ ⑥ ⑦ ⑧ `.0-255.0-255`

0 0 → S_1 → 52.0.0.0 to 52.63.255.255

0 1 → S_2 → 52.64.0.0 to 52.127.255.255

1 0 → S_3 → 52.128.0.0 to 52.191.255.255

1 1 → S_4 → 52.192.0.0 to 52.255.255.255

52.192.0.0 — Subnet id's

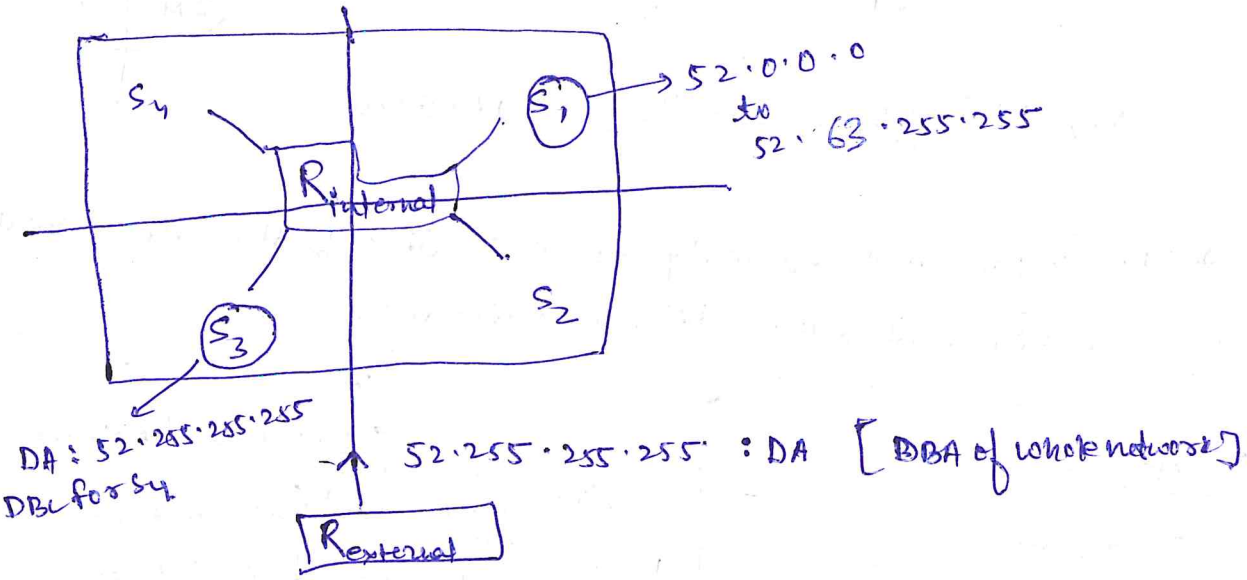
52.255.255.255 — DBA of Subnet id's

* No. of hosts possible in each subnet is again

* $2^{\text{host bits}} - 2$ ⇒ In last Ex→ $2^{22} - 2$ for each subnet.

Que: 52.255.255.255 ⟶ Direct broadcast of NW or S_4

52.0.0.0 ⟶ Nw-IP of NW or S_1



* If a packet coming from internal network.

52.255.255.255 then the packet is treated as broadcast address of S_4

* If a packet is coming from outside network then it is treated as broadcast address for whole network.

Subnet mask : 32-bit

In the subnet mask all the network and subnet bits are 1, host bits are zero.

Ex-1 $255.192.0.0 = /10$ → Another representation of subnet mask.

8 | 2 | 000000 . 0 . 0

net | Subnet

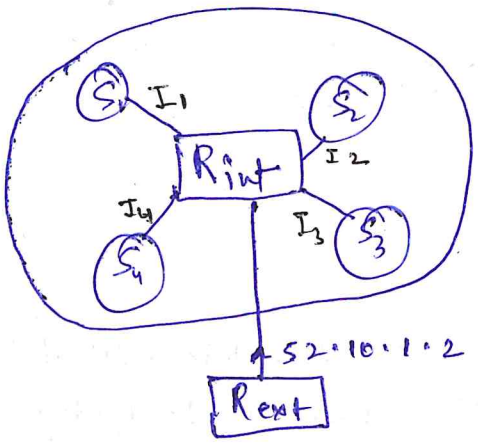
* Subnet mask is calculated at the time of subnetting itself.

Consider previous figure and Ex:-

Rint: ① [Subnet mask : 255.192.0.0]

② if:

- 52.0 – 52.63 → S_1 (I_1)
- 52.64 – 52.127 → S_2 (I_2)
- 52.128 – 52.191 → S_3 (I_3)
- 52.192 – 52.255 → S_4 (I_4)



If Rint receives the packet with these IP. It will forward to corresponding interface.

③ else, if the ip dresnot belong this range then use mask 0.0.0.0 and send to R_{ext}

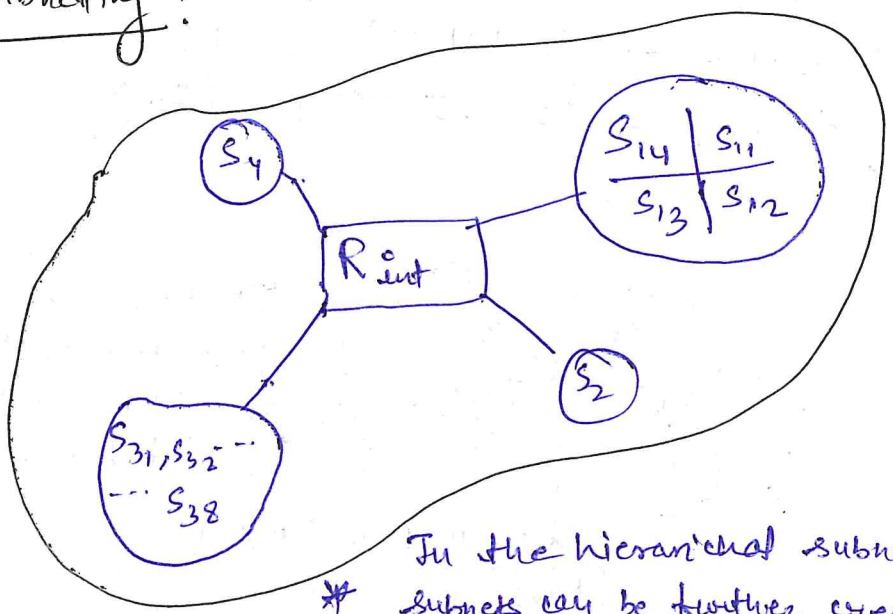
Ex→ If R_{ext} send the packet with IP 52.10.1.2, then R_{int} will perform masking.

Performing logical AND with subnet mask no. and host address will give you subnet id.

```
  52 . 0000 1010 . 1 . 2 .
 255 . 1111 0000 . 0 . 0
-----------------------------
AND   52 . 0 . 0 . 0
```

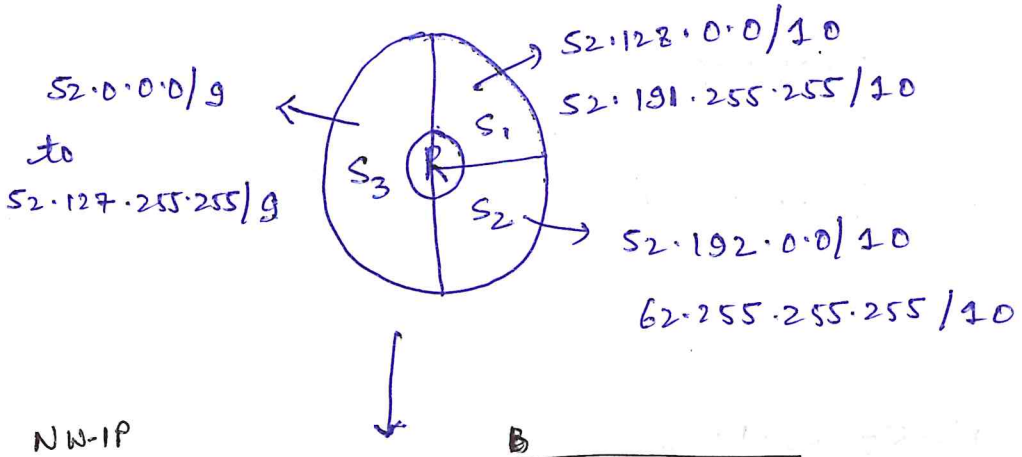
* By getting the subnet id R_{int} will forward to the corresponding subnetwork.

Hierarchical - Subnetting :



* In the hierarchical subnetting subnets can be further created in subnets.

Subnetting : different Sized - Si's



NW-IP

52.0.0.0/8 to 52.255.255.255/8

/8 → Subnet mask of whole network.

255.0.0.0

Dividing network id [we want to give half of the address to S_2.

Net id → 52.0.0.0

52.0 ——————.0-255.0-255

0 → S_3 [Half of the address]

10 → S_1 [Quarter of the address]

11 → S_2 [Quarter of the address]

S_0, S_3 ⇒ 52.0.0.0/9 to 52.127.255.255/9

↳ mask no. is 9 because we are using 8 net bits and 1 Subnet bit

S_1 = 52.128.0.0/10 to 52.191.255.255/10

↳ mask no. is 10 because we are using 8 net bits 2 subnet bits.

S_3 = 52.192.0.0/10 to 52.255.255.255/10

↳ mask no. is 10 because we are using 8 net bits and 2 Subnet bits.

R	Mask	Subnet id	Interface / Subnetwork.
(1)	/9	52.0.0.0	S_3
(2)	/10	52.128.0.0	S_1
(3)	/10	52.191.0.0	S_2.

E.g. D.A. → (a) 52.62.1.2.

↓

Mask with /9 [means take AND with mask and Address]

I get 52.0.0.0 ⇒ S_3 ✓.

(b) 52.161.1.2

↓

Mask with first entry /9 ✗ does not get S_3

mask with second entry /10 ⇒ S_1

↳ 52.128.0.0 ✓.

(c) 52.200.1.2.

↓

Mask with first entry /9 ✗ doesnot get S_3

Mask with second entry /10 ✗ does not get S_1

mask with third entry /10 ✓ ⇒ S_2.

This is how internal router do masking.

[CIDR] Classless - Inter Domain Routing (1993 - IETF)

* The task force created in 1993 started getting problems in classfull addressing.

* In 1993 people started using classless.

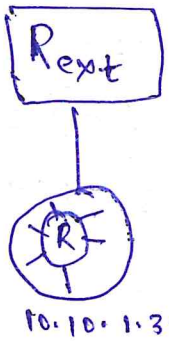
* Assign any Block of IP addresses to any ISP/company/inst
  ↳ IANA

* IANA assigns consecutive Ip addresses upto the number you want. Let's we want 2^{16} addr.

  Eg $a.b.c.d + 2^{16}$
  - $a.b.c.d$ ↓ first addr
  - 2^{16} ↓ consecutive addr

* Let's take an Ex: we want 32 IP address for company.

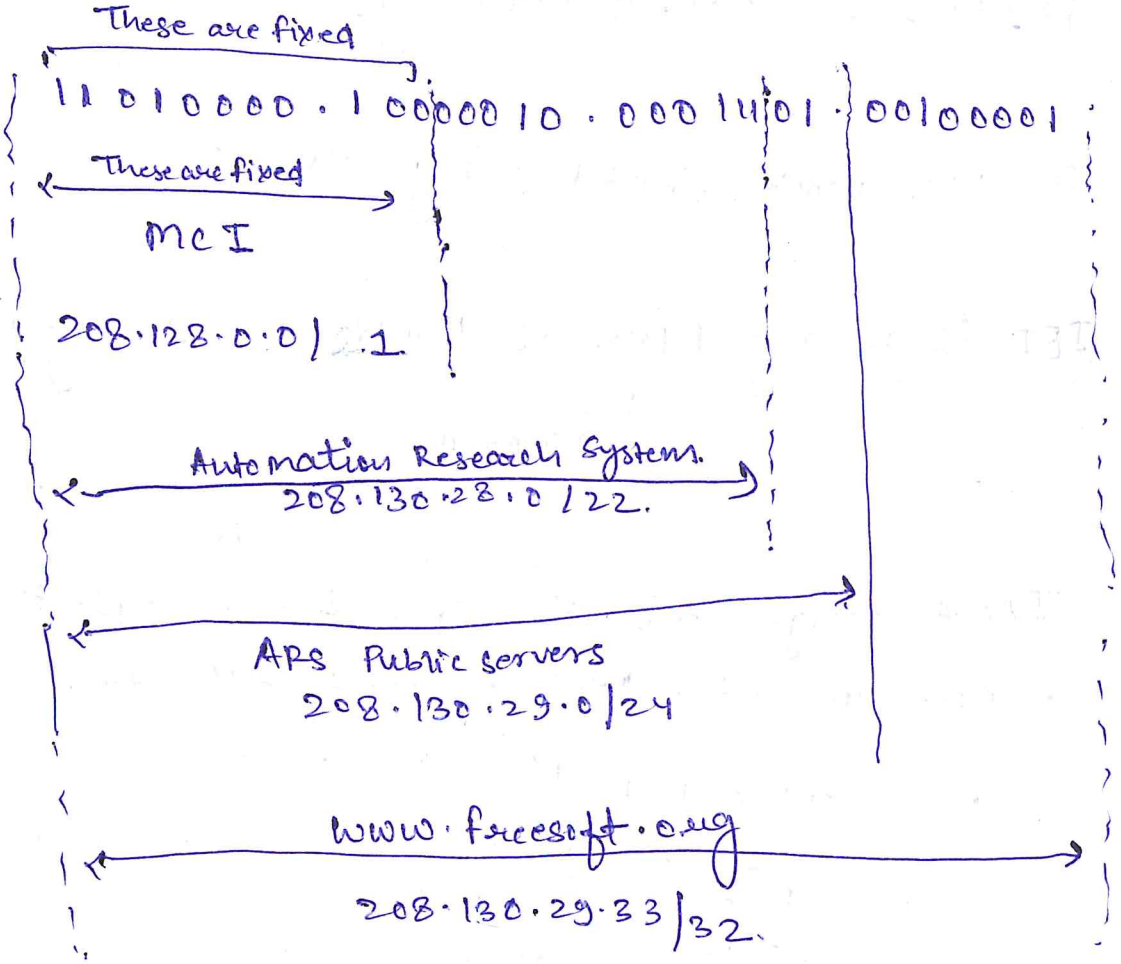
  IANA: → Assigned a block.
  - $10.10.1.32/27$ → This is called block address
  - $10.10.1.63/27$ → This is called BC-IP



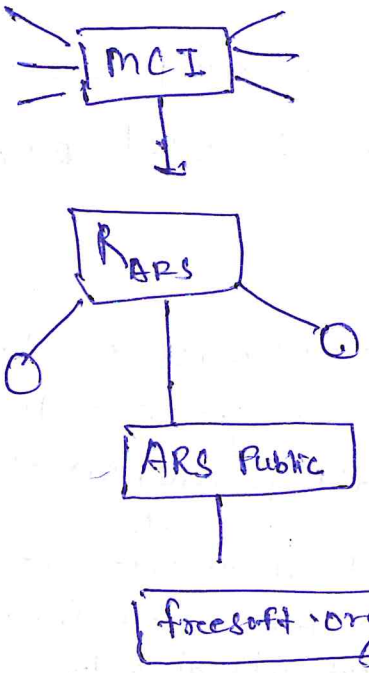
* Router ext will mask the host address
  ∴ If we get the 10.10.1.32
  It will forward to this network.
  OR
  if it does not get 10.10.1.32 and get suppose 10.10.1.90 then Rext will discard.

Late 1990's

IANA Gave 208.130.29.33



It is a hierarchical subnetting.



IANA → MCI → ARS → PS →
2 mill 2^{10} 2^{8}

IANA assigns to MCI
MCI assigns to ARS some addresses
ARS assigns to PS some addresses.

freesoft.org → only one address assign because /32 is mask.

a.b.c.d /32 → 2^0 → only one host we can assign

a.b.c.d /31 → 2^1 → Two addr possible

a.b.c.d /27 → 2^5 → 32 addr possible

Problems:

Que: Consider three machines M, N and P with IP addresses 100.10.5.2, 100.10.5.5 and 100.10.5.6 respectively. The subnet mask is set to 255.255.255.252 for all three machines. which one of the following is true?

(A) M, N and P all belong to the same subnet

(B) only N and P belong to the same subnet.

(C) M, N and P belong to three different subnet.

(D) only M and N belong to the same subnet.

Ans: 255.255.255.11111100 = /30

For solving this question we have to apply AND between subnet mask and Ip address to get the subnet Id's.

M ⇒ 100.10.5.00000010
255.255.255.11111100
Subnet ⇒ 100.10.5.0

N ⇒ 100.10.5.5
100.10.5.00000101
255.255.255.11111100
Subnet ⇒ 100.10.5.4

P => 100.10.5.00000120
255.255.255.11111100

Subnetid => 100.10.5.4

=> Therefore, N and P belong to the same subnet.

Que: Consider the following routing table at an IP router for each IP address in Group-I identify the correct choice of the next hop from the Group-II using the entries from the routing table above.

List - I	List - II
A. 128.96.171.92.	1. Interface 0
B. 128.96.167.151	2. Interface 1
C. 128.96.163.121	3. R_2.
D. 128.96.165.121.	4. R_3
	5. R_4.

Network No.	Netmask	Next hop
128.96.170.0	255.255.254.0	Interface 0
128.96.168.0	255.255.254.0	Interface 1
128.96.166.0	255.255.254.0	R_2
128.96.164.0	255.255.254.0	R_3
0.0.0.0	Default	R_4

Codes:

	A	B	C	D
(a)	1	3	5	4
(b)	1	4	2	5
(c)	2	3	4	5
(d)	2	3	5	4

Ans: for each Ip address in List-1 we have to find the correct choice of next hop.

A ⇒ We will perform AND between

128.96.171.92.
255.255.254.0.

128.96.170.0

So, A will go to interface 0.

c and d options cannot be the answer.

B ⇒

128.96.167.151
255.255.254.0

128.96.166.0

B will go to R_2.

Hence answer is a.

167 ⇒ [XXXXXX]1
254 ⇒ 1111110
[XXXXXX]0
166

Short trick.

Que: In the network 200.10.11.144/27, the fourth octet (in decimal) of the last Ip address of the network which can be assigned to host is —

(a) 158
(b) 255
(c) 222
(d) 223

Ans: 32 - 27 = 5 host bits.

200.10.11.144

last octet =>

=> 1 0 0 1 0 0 0 0 = 144

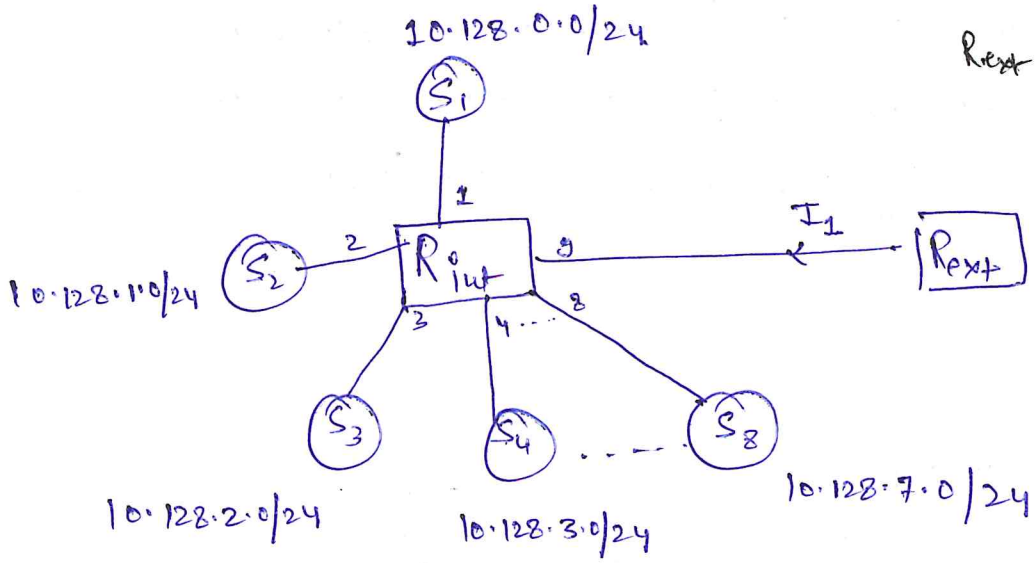
network

So, Max 1 0 0 1 1 1 1 1 → Largest host

So, 158 is the last octet.

Supernetting:-

In supernetting we combine two networks logically.



Rext => routing table.

M	NW	Interface
/24	10.128.0.0	I_1
/24	---1.0	I_1
/24	----	I_1

Rext will see the corresponding mask no. and network address. Forward it to interface.

Rint table:

M	NW	Interface
/24	10.128.0.0	1
/24	10.128.1.0	2
⋮ /24		⋮ 8
/0	Default	9

* The routing table of Rext is becoming too large because of the subnets are more. The size of Rext routing table is not sufficient to handle this big data. If n etries their we need O(n) time to decide which interface we need to send.

* In supernetting, we are logically combining the subnetworks with the objective of reducing the size of external routing table.

Example -

These are same in all

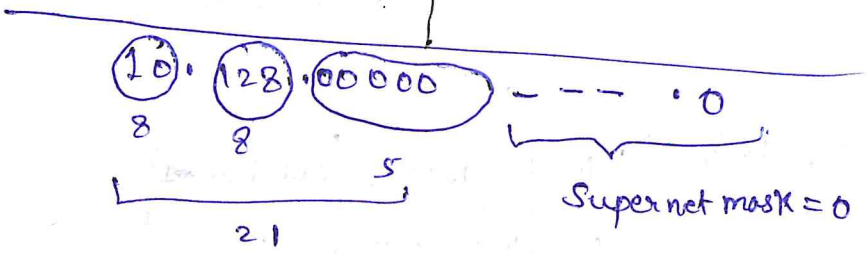
S_1 : 10.128.0.0/24. → 10.128.0000 0000.0

S_2 : 10.128.1.0/24 → 10.128.0000 0001.0

S_3 : 10.128.1.0/24 → 10.128.0000 0010.0

⋮

S_8 : 10.128.7.0/24



Here we are changing our mask number from 24 to 21 [Because upto 21 all bits are same] for all subnets.

Now, In Rest table.

NW	NW	Interface
/24	---	I_1
/24	---	I_1
⋮		
/24	---	I_1

Previous table

→

NW IP	mask	Interface
10.128.0.0.	/21	I_1

We are replacing these 8 entries with single entry.

E.g. Rext get 10.128.7.6 as the destination address.

= Rext will mask with 21.

10.128.00000111.00000110
255.255.11111000.00000000

10.128.0.0

Hence, Rext will send this packet to Interface 1.

E.g DA: 10.128.3.12 ⇒ Rext get this packet

= 10.128.00000011.00001100
255.255.11111000.00000000

10.128.0.0 ⇒ Packet sent to Interface 1.

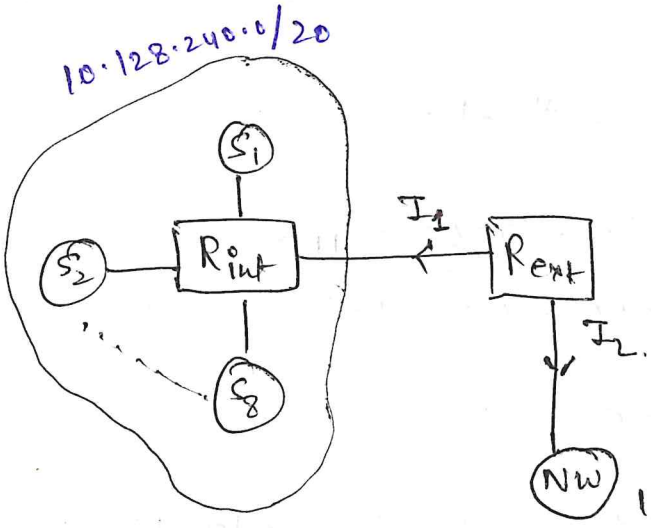
Example – 2

First 4 bits are same in all so keep them same.

S_1	10.128.247.0	10.128.11110111.0
S_2	10.128.248.0	10.128.11111000.0
S_3	10.128.249.0	10.128.11111001.0
⋮		⋮
S_8	10.128.254.0	10.128.11111110.0

10.128.240.0/20 ⇐ 10.128.1111----.0

Supernet mask = /20 ⇒ 1



Routing Table

10.128.240.0/20	, I_1
10.128.241.0/8	, I_2

Eg: DA: 10.128.241.3 → 10.128.1111 0000.0 → I_1 (INCORRECT)

When mask with /20

* It should have been sent through I_2 : this would be more appropiate. Just I_1 entry is above than I_2. It has sent to I_1.

This is incorrect supernetting.

Necessary condition to perform correct supernetting.

S_1 : 10.128.247.0/24

S_2 : 10.128.248.0/24

S_3 : 10.128.249.0/24 ⇒ we got Supernet mask /21 for this.

⋮

S_8 : 10.128.254.0/24

The lowest Subnet address ⇒ 10.128.247.0

10.128.1111 0111 ✗

21 0000 ✓

→ After 21 bits of mask. everything should be zero in the lowest subnet address.

→ This way you will get the correct supernetting.

Necessary conditions: -

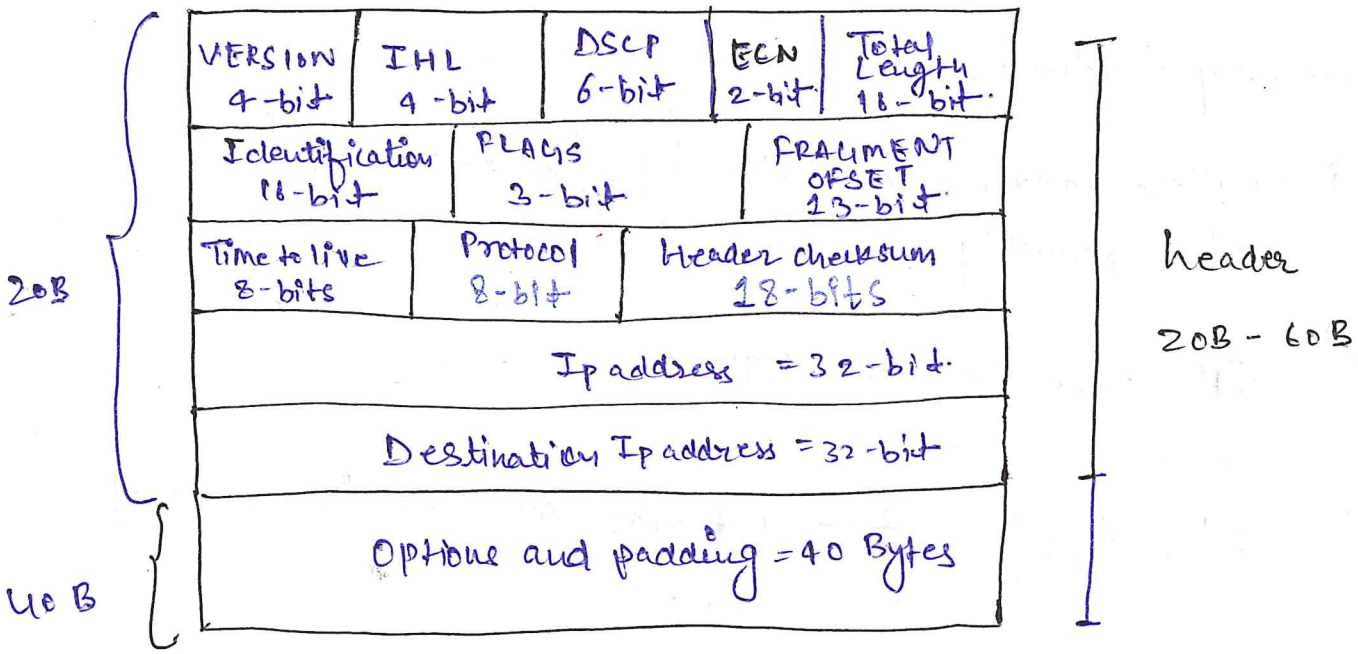
(1) Smallest address divisible by /x that is obtained post supernetting

(2) IPs must be in sequence

(3) $2^1, 2^2, 2^3$ ← Networks combined in powers of 2.

IPV4 : Packet format

Header :



Version → version tells us which Ip version you are using

0100 = IPV4

0110 = IPV6

IHL : Internet header length → It tells what is the length of the header

20B is the mandatory. [Min HL]

Concept → what ever the value in IHL field multiplied by 4 will give you header length.

→ 1111 = 15 × 4 = 60B (IHL)

→ 1010 = 10 × 4 = 40B

IHL minimum field = 0100 = 5 × 4 = 20 B

* If your header length → 20 + 31 B = 51 B
(20 ↓ Mandatory)

↳ 51 B is not divisible by 4

→ Add 1 B padding = 52/4 = 13 ⇒ (IHL = 13) ✓

③ DSCP (Differentiated - services code point) - wiki

↳ Types of service { VoIP & videochat.

Ex→ 101 110 ⇒ Expedited forwarding [Forward as soon as possible]

000 000 ⇒ Best effort

④ ECN (Explicit congestion Notification)

→ often combined with Type of service field

→ To notify that there is congestion.

⑤ Total length (16-bit) → It will give you the size of whole packet.

→ 2^{16} = 64 KB (Max size: header + data)

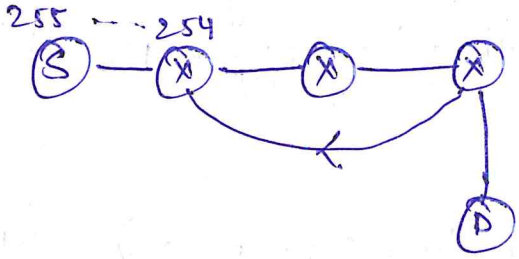
→ 20 B: min size [Here let suppose we don't have any data.

IP packet

(6) Source and Dest Ip : 32-bit each

(7) Time to live : 8-bit field.

→ Count on numbers hops you have.



⇒ If a packet is continue looping will create a lot of congestion.

Max value (8-bit) = 255

→ Max no. of hops = 255 = 11111111

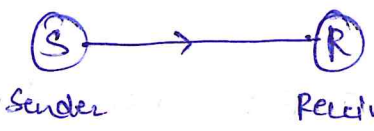
→ In each hop value decrease by 1.

→ If your value is zero your packet will be dropped.

→ changed by the routers in between.

TERM:

DATAGRAM : Packet of an unreliable service.



In un-reliable service we don't get the acknowledgment for the sending packet.

(8) Protocol →

→ Protocol of the data part (TCP/ UDP/ ...)

→ List of IP- protocols numbers (WIKI)

→ ICMP/ IGMP/ UDP/ TCP } → Transport layer protocol.

It tells us what protocol is used in the data field of IP packet.

(9) Header - Checksum:

→ data checksum are used

→ Header checksum itself filled with zeros to compute the checksum } → It is computed by sender

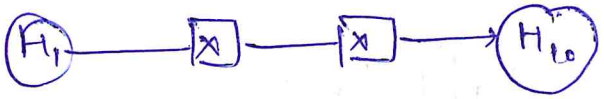
→ Router: TTL changed on every router ⇒ checksum also changed. + other field may change

So, for solution router again re-computes the checksum.

(10) options: (ISP and network managers)

It is used by the ISP providers or network managers.

(a) Route-recording: —



SA = H_1

DA = H_{10} Route recording is used for recording the route.

Like → | R_1 | R_2 |
32 Ip 32 Ip

Options = max 40B.

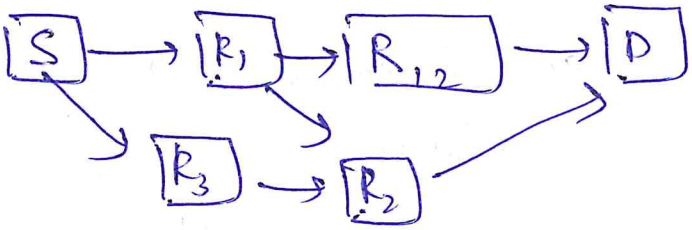
{ 10 Ip address in between route can be stored.

→ But actually we are able to store max 9 Ip addresses because 2B are reserved for future.

So, it is $\frac{38B}{4B} = 9$ IP

(2) Source-routing option:

(a) Strict Source routing :-



Means always take the mention path.

| R_1 | R_{12} | D | fixed cannot change

(b) Loose source routing :-

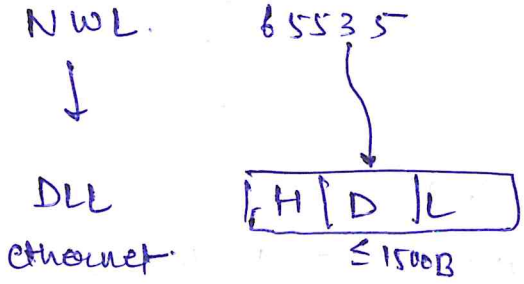
| R_1 | R_2 | D |

R_{12} D → we can also take alternate paths

Fragmentation →

fragmentation means break something up into smaller pieces.

The size of Network layer packet is 65535 Bytes.



The packet is placed in frame.

So, the whole packet we can not transfer directly we have to break the packet.

MTU (maximum transmission unit)

MTU for ethernet is 1500B.

→ Every network has its MTU field.

ID (16-bit) → Every datagram has unique ID which is given in Network layer.

FLAGS (3-bit). First bit always zero not used.

0 Df Mf

Df means do not fragment.

DF = 1 (It denotes it is packet)

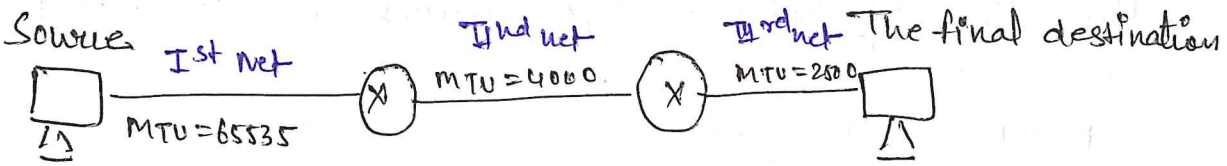
DF = 0 (It is a fragment)

MF (More fragments)

MF = 1. (It is an intermediate fragment)

MF = 0 (It is a last fragment)

Ex →



Original Packet: -

ID	offset	More frag	Pay load
578	0	0	10,000

20B

= 10,020 B

for first network we need not to fragment the packet because MTU > Packet size.

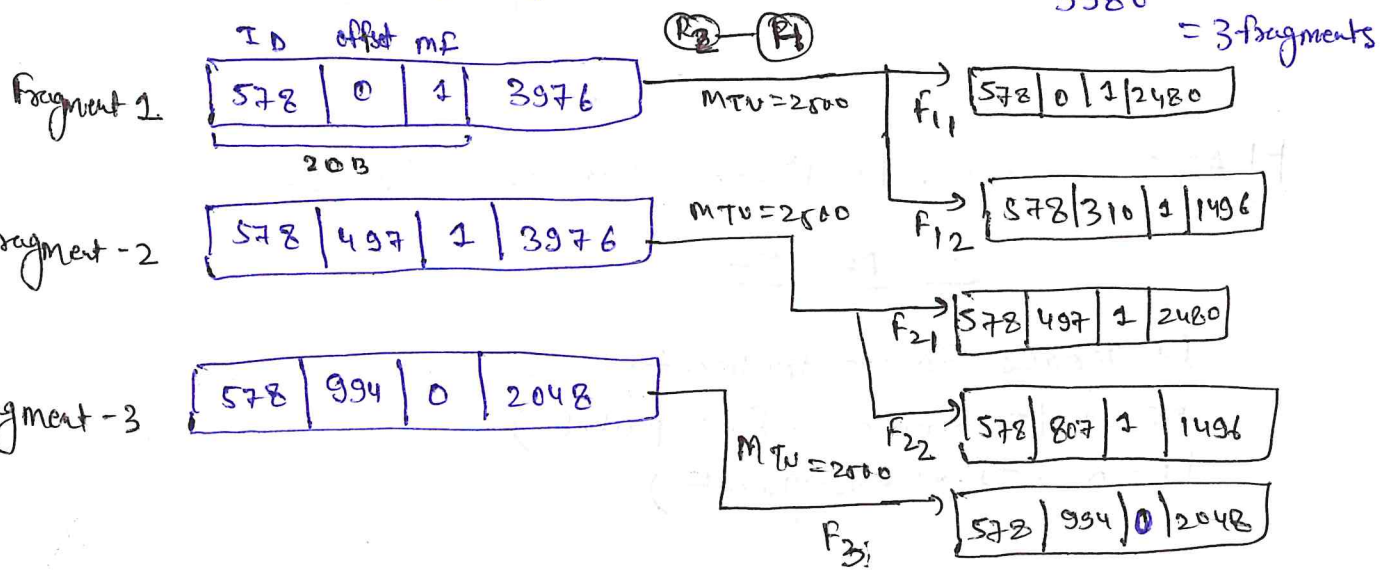
R_1 —(2nd)— R_2 In 2nd Network MTU is = 4000B

MTU < Packet size.

So, we need to do fragment

* Number of fragments for network 2 = $\frac{10,000}{3980}$ = 2.51

= 3 fragments



Fragment 1 offset = 0 because start with first byte.

fragment 2 offset = 497 because start with $\frac{3976}{8}$ = 497th byte

fragment 3 offset = 994 because start with $\frac{3976+3976}{8}$ = 994th byte

** Note → Now question is why we are using 3976 bytes instead of 3980 in starting?

Ans: * Because our payload should be always divisible by 8

3980 → not divisible by 8

3976 → divisible by 8.

*

→ Fragment 1 further divided into fragment 11 and fragment 12

3976 > MTU

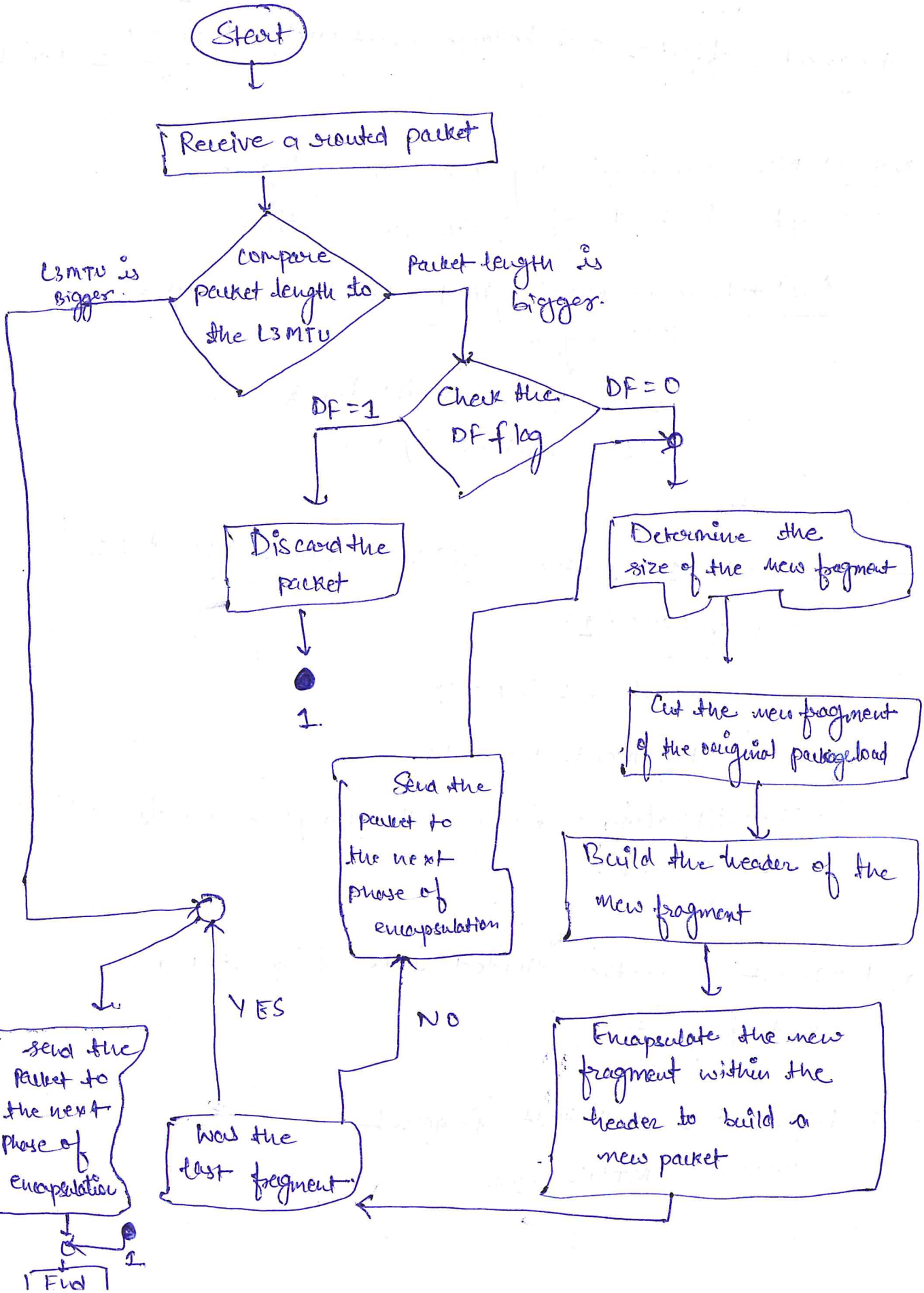
2 fragments possible.

Identification number will always same for all fragments.

→ Fragment 2 further divided into fragment 21 and fragment 22

→ fragment 3 will not fragment because MTU > 2048

Fragmentation Algo (IPV4)



Que. Consider an IP packet with a length of 4500 bytes that includes a 20-byte IPv4 header and 40-byte TCP header. The packet is forwarded to an IPV4 router that supports a Maximum Transmission Unit (MTU) of 600 bytes. Assume that the length of the IP header in all the outgoing fragments of this packet is 20 bytes. Assume that the fragmentation offset value stored in the first fragment is 0. The fragmentation offset value stored in the third fragment is ____.

600 - 20 = 580 B data.

Ans:

	Offset	MF	Data.
F1:	0	1	576 B → It should be divisible by 8
F2:	72 (576/8)	1	576 B
F3:	144	1	576 B

the fragment should be divisible by 8

The fragment offset for third fragment is 144.

Que The maximum number of IPV4 router address addresses that can be listed in the record route (RR) option field of an IPV4 header is ——.

Ans:- 40 B → optional

2 B → Reserved

38 B left

In $\frac{38\ B}{4\ B}$ = 9 IP addresses possible

Que: An IP datagram of size 1000 bytes arrive at the router. The router has to forward this packet on a link whose MTU (maximum transmission unit) is 100 bytes. Assume that the size of the IP header is 20 bytes. The number of fragments that the IP datagram will be divided into for transmission is: ?

Ans: Data = 80 B Header = 20 B No. of frag = $\frac{980}{80}$ = 12.5

F1 80 B → 80 divisible by 8 no problem.

	offset	
F1	0	80 B
f2	10	80 B
F3	20	80 B
	⋮	

12.5 = 13 fragments ✓.

Que: Which of the following fields of an IP header is not modified by a typical IP router?

(a) checksum (b) Source address (c) Time to live (TTL) (d) Length.

Ans → b Checksum → modified

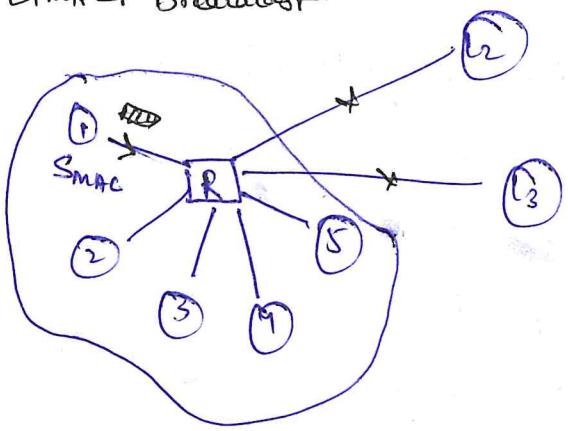
Source address → not modified

TTL → modified

Length → modified

IPV4: Broadcasting

(1) Limited Broadcast.



→ To communicate in the same LAN we use Ethernet protocol

SA: IP_1

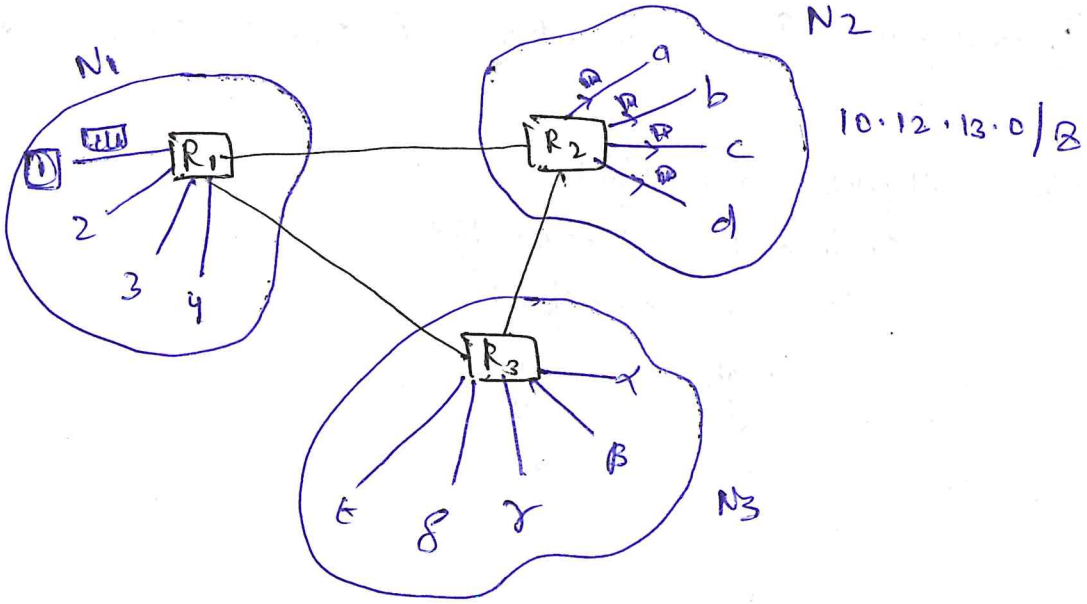
DA: 255.255.255.255. → Router after seeing this will not transfer to the other LAN's.

In limited broadcast the destination address contains all one's.

DLL: S_{MAC}

D_{MAC}: all F's] → 12 digit hexadecimal number 48 bit. All F's means broadcasting.

5) Directed Broadcast : In directed broadcast we are sending a broadcast packet in the other network.



SA: Ip address of host 1 from N1

DA: 10.12.13.255 → DBC - IP for N2.

Packet reached to R_2

R_2 looks the destination address

So, R_2 in DLL sets to all F's

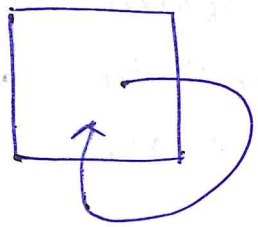
Now, all hosts of N_2 can get the packet

Local host (Loopback) / Localhost.

→ 127.0.0.1 [Reserved IP address]

* Using loopback address we can test the connection with our own computer.

* It is a delay of sending a packet to node to itself

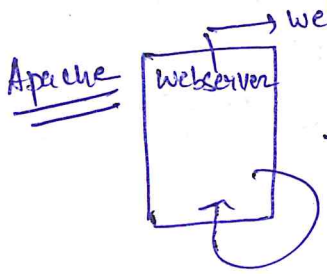


→ Ping (or) http://localhost

→ Debugging | Testing

* It is used to debug or test the network.

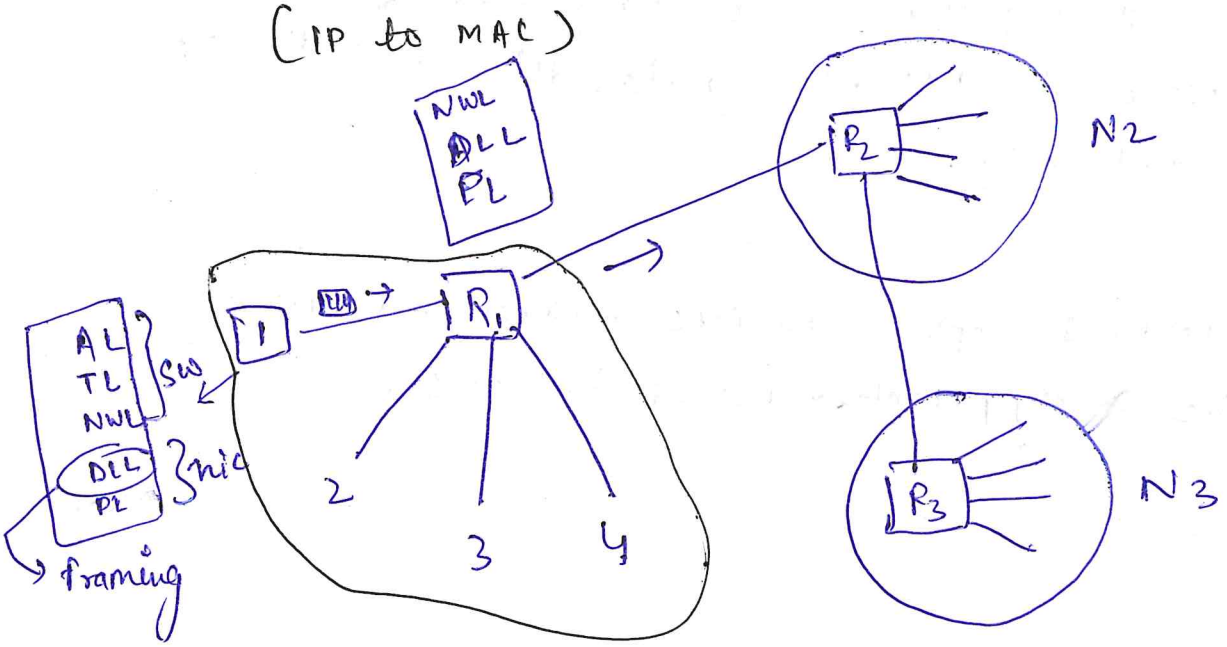
* localhost is also used to test the webserver.



→ It is used to test the server on our own computer.

Address Resolution Protocol (ARP)

(IP to MAC)



→ When you send a packet your host works on several layers before sending the packet.

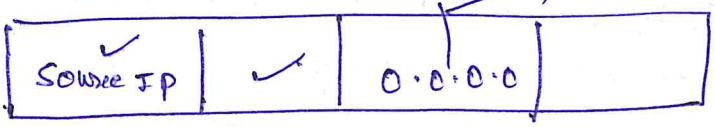
→ Now, when Host 1 sends the packet it sends the packet with Ip address of destination and MAC address of R_1 is unknown to H1,

→ ARP → Address resolution protocol helps us to resolve the MAC address of the destination.

It works on req and reply scheme.

Host 1 wants the mac address of destination router R_1

Initially don't know so filling zero



ARP request packet

This is a broadcast packet

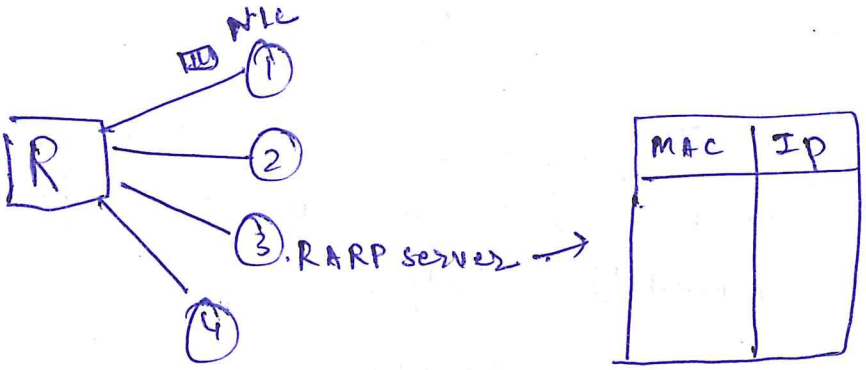
* This request packet will be transfer to all the hosts in the LAN. Whoever has this destination IP will reply.

* So, Router 1 replies with MAC address and updates on the reply packet and send to host 1.

RARP: MAC to IP

Reverse Address resolution Protocol

→ MAC fixed
IP : variable



* RARP is used to get the IP address

* Host 1 wants an IP address, so it broadcasts by using all F's in the MAC address in the data link layer.

The RARP server replies with the response if it has the entry of MAC in the table.

If it is a new computer then RARP server creates the entry of the new mac address and sends the new IP address.

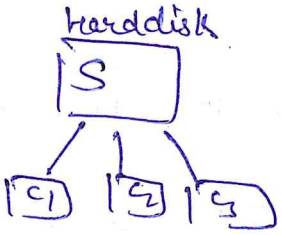
BOOTP (1985) [MAC - IP]

→ Bootstrap protocol.

→ Application layer protocol

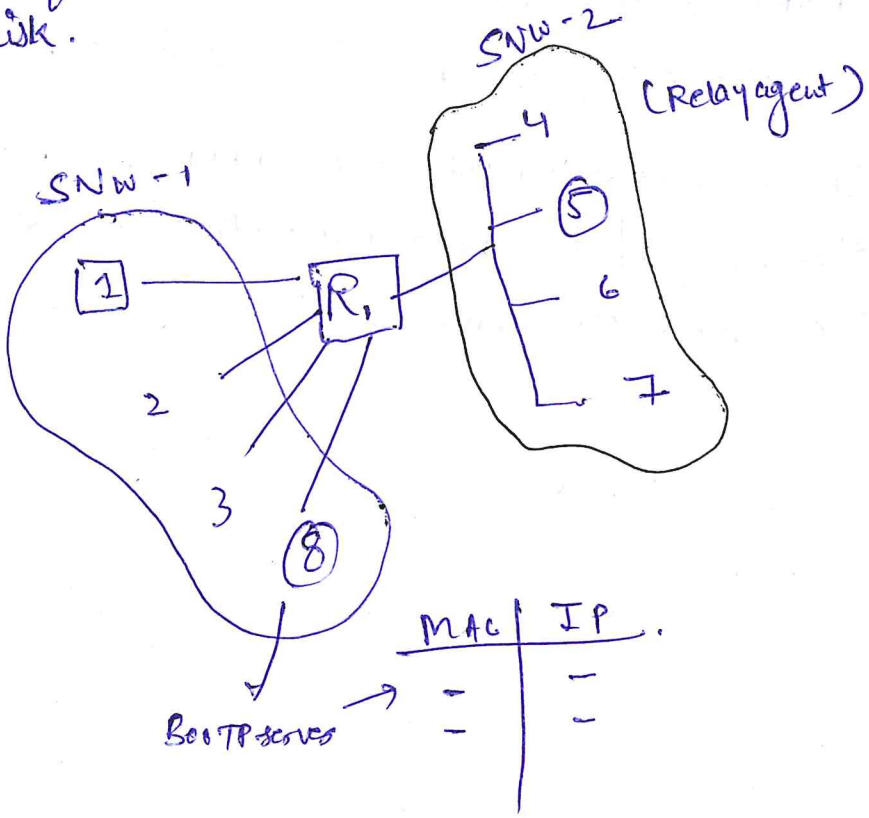
→ Created for Linux thin clients

Thin clients mean: The systems who have less memory or power. They rely on server for the computational help.



* BOOTP is not the popular protocol now-a-days

* BOOTP protocol are also used for getting the IP address for the thin clients who have no memory or disk.



It is also like RARP request where you the host request for IP by sending a broadcast packet. The BOOTP server will respond with the reply.

* The concept of relaying help us to get only one BOOTP server amongst all subnetworks.

Ex→ In the subnetwork 2 Host 5 wants an IP address. when host 5 send the broadcast packet to all the hosts. The relay agent is the special node which knows the address of BOOTP server.

Relay agent sends the unicast request to BOOTP server and BOOTP server will reply with the IP address afterwards relay agent send the IP to host 5.

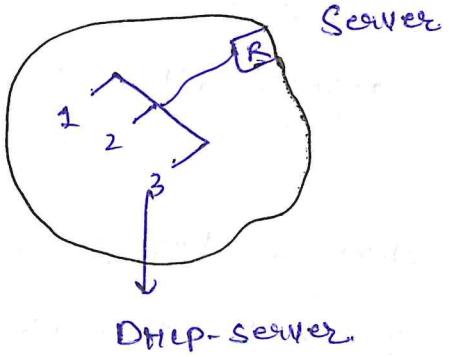
* The drawback in RARP/BOOTP is they have static table on the server.

The static table contain the IP address and mac address. The IP and mac are fixed

→ In the static table IP's are fixed so that we can not reuse them. Let suppose we have 10 systems and we are only using 3 system the rest of 7 systems IP's are not usable at that instant of time.

DHCP (Dynamic host configuration Protocol)

→ Most widely used.
→ dynamic + static entries
→ relaying (Like in BOOTP)
→ lease time



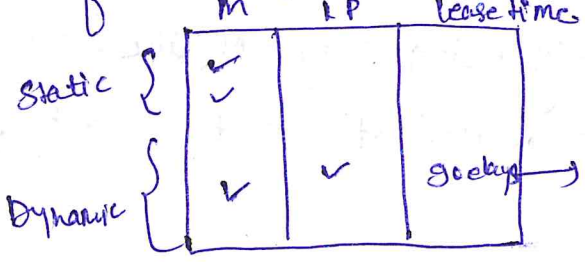
* In the DHCP there is also a concept of relaying like in BOOTP.

* Here we have dynamic as well as static entries in the DHCP server.

Static entries are for those systems who always connect with the internet like server [That are always on]

* Set of the IP addresses are available that are non-assigned. When a new computer connect DHCP server assigned one IP address amongs non-assigned IP address.

There is a concept of lease time when DHCP server assigned an address. It also assigned the lease time that is valid for that particular IP address.



Once the lease time is over. We need to again assign the IP address.

Private vs Public IP:

* We have only 2^{32} public IP addresses. To save the public IP addresses we use the concept of private IP- address.

* ISP has only Public Ip addresses which is limited to some number

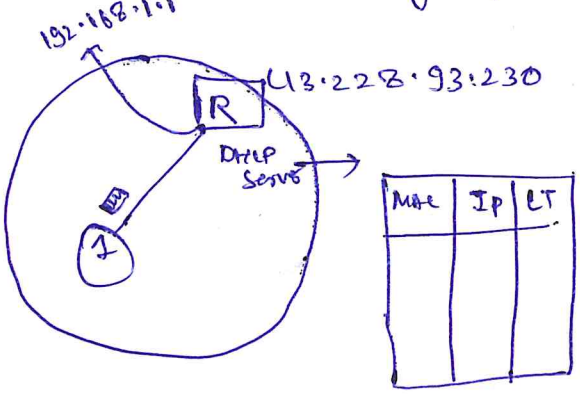
Private IP address:

10.0.0.0 to 10.255.255.255
172.16.0.0 to 172.31.255.255 [IANA]
192.168.0.0 to 192.168.255.255

These are the range of private IP address.

* We can effectively utilize the public IP addresses with the help of private IP addresses.

The public address will assign to router and rest of the system inside LAN will assign private addresses.



The host I is assigned with private IP address with the lease time by the router or DHCP server.

Host 1 sends the packet by

SA: 0104 private address Assigned by DHCP server

DA: 192.168.1.1 (Router internal address)

When router further sending the packet outside LAN.

SA: 43.228.93.230 (Public ~~Google~~ Address)

DA: Google Address.

* For getting IP address from DHCP server.

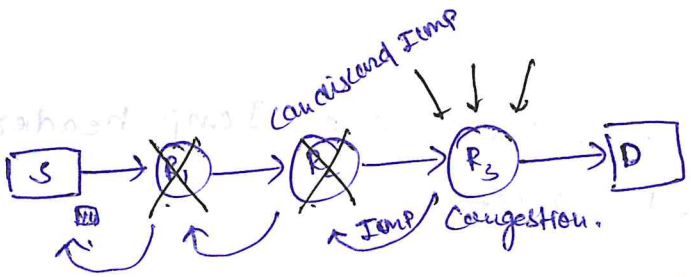
Sender sends. SA: 0.0.0.0 + MAC address → Broadcast packet.

* The router that converts your private address to public address while sending outside are called the NAT router. [Network Address Translation].

ICMP (Internet control message protocol.)

Internet Protocol (IP) → Unreliable → no ACK.
Datagram

Ex-1



packet crossed R_1, R_2. . At R_3 there is congestion

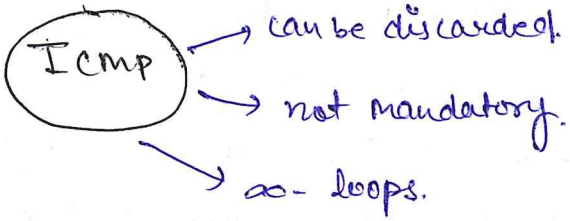
The Router R_3 discards the packet because of the congestion.

→ The Router 3 drops the packet so it will inform the source using Icmp. [sending Icmp is not mandatory]

IP + Icmp ⇒ Unreliable because

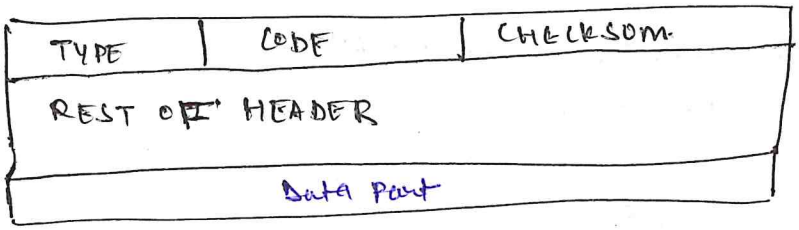
Icmp packet can sometime be discarded.

↳ TCP → reliable.

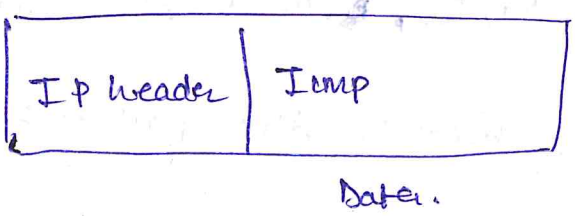


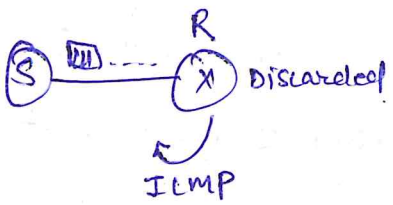
* for the discarded Icmp packets we don't send Icmp again

ICMP header format



* The Icmp packet sits inside the data part of IP packet



* 

* Icmp packet has header + Data part

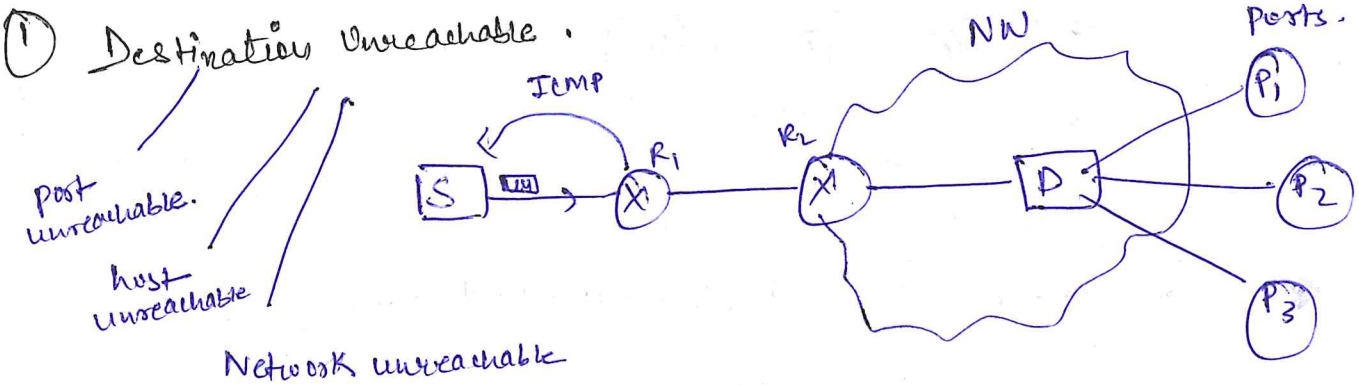
* In the data part of ICMP packet we put the header of IP packet + 8 Bytes of data of IP packet

↓

This ICMP packet when reaches the sender. By looking at header and data part of ICMP packet. Sender get to know about which packet has been lost.

Types and code for Icmp: WIKI.

(1) Destination Unreachable.



(a) Port unreachable → The R_2 can not send the packet to port in D. So port unreachable.

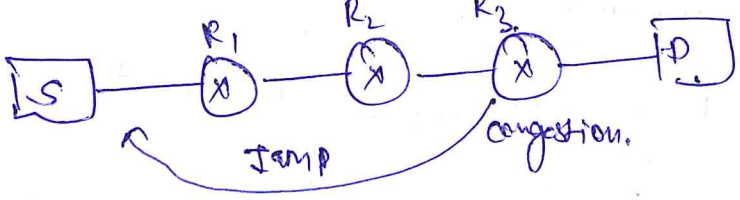
(b) Host unreachable → Host can be unreachable.

(c) Network → R_1 can not reach to NW.

* Ports helps us to send the packet in the specific program or process on the destination.

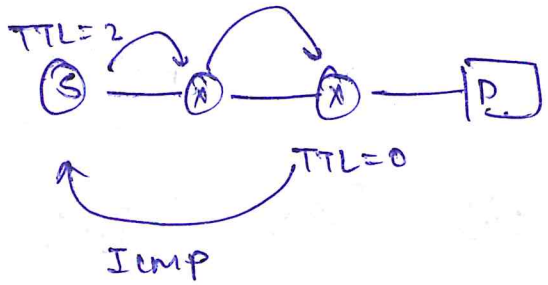
→ There are various types and codes are available in Icmp.

(2) Source quench: Type 4



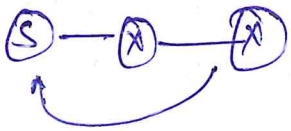
The Router 3 sends the ICMP packet to the source that stop sending the packets.

③ Time exceeded :



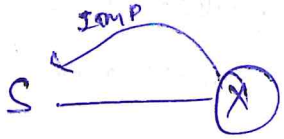
* TTL = 0 so send time exceeded message.

④ Parameter Problem : —



There is problem in Ip header. So router send an Icmp packet saying that header has some problem.

⑤ Router advertisement and solicitation :-



The router advertisement : I am the router that everyone should use.

Ping and Traceroute :

Ping google.com → echo request.

Try this on terminal ←

64 bytes from 172.217.167.174 icmp seq=0 ttl=53 time=19.942 ms

↓

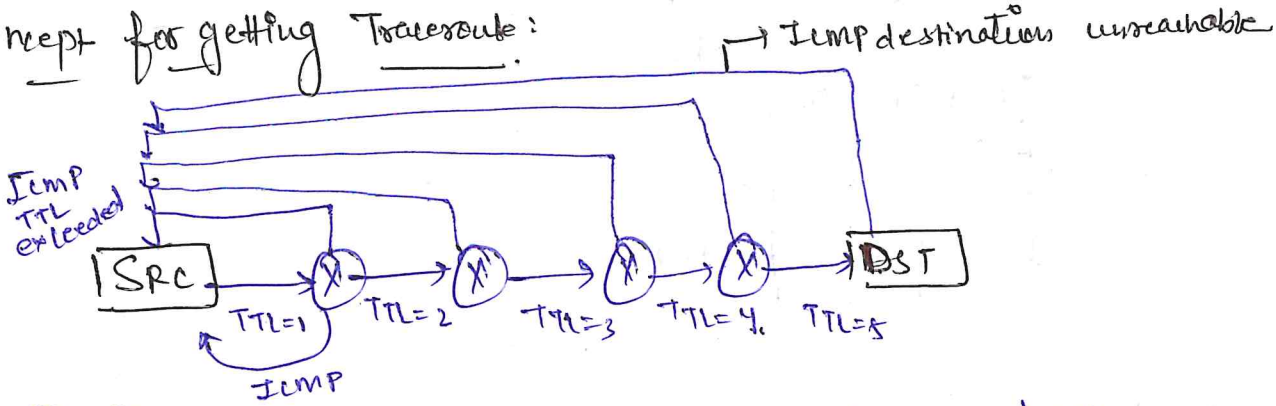
echo Reply from ping command.

* Ping helps you to know everything is working perfectly.
Test / Debug network.

Traceroute: Traceroute amazon.com

By using traceroute you will get to know the IP address of all the computer/router in the network in between amazon.com.

Concept for getting Traceroute:



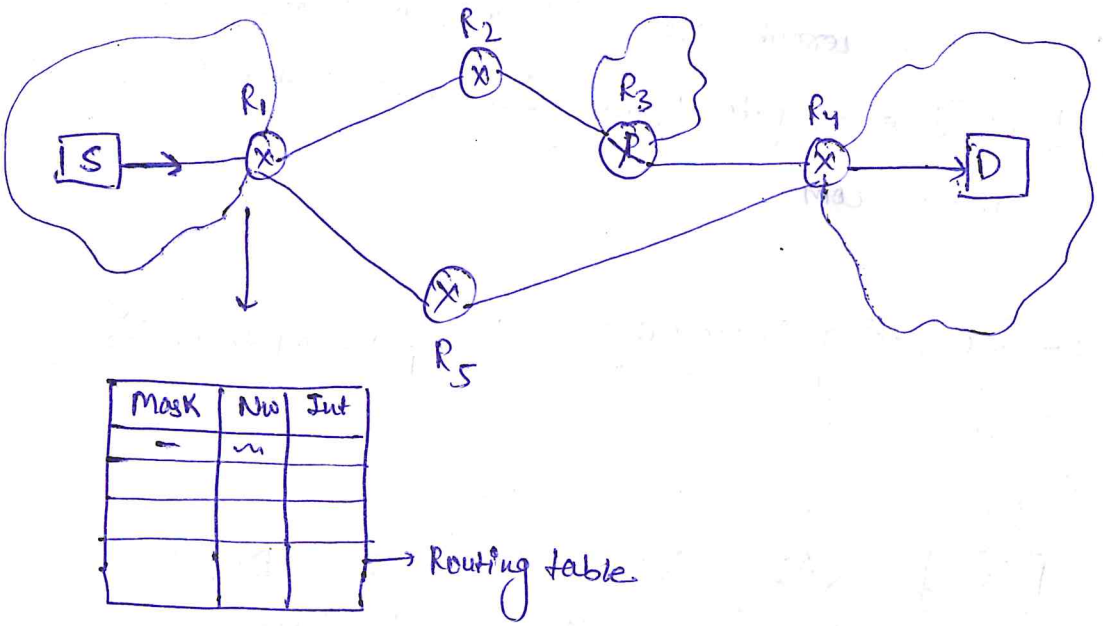
(a) First we assign TTL=1 so packet goes from source to first router and we get the address of first router, because first router send the Icmp packet with message Icmp TTL exceeded

(b) Second we assign TTL=2 we get the address of 2nd router.

⋮

Like this we get the IP address of whole routers in the route.

Routing :



* After seeing the destination address every router has to make this decision what is the next hop device.

* Routing is all about entries in routing table. How the packet will be guided by the routers to reach the destination.

* Routers interact with each other to fill the routing tables. Every router has the routing table.

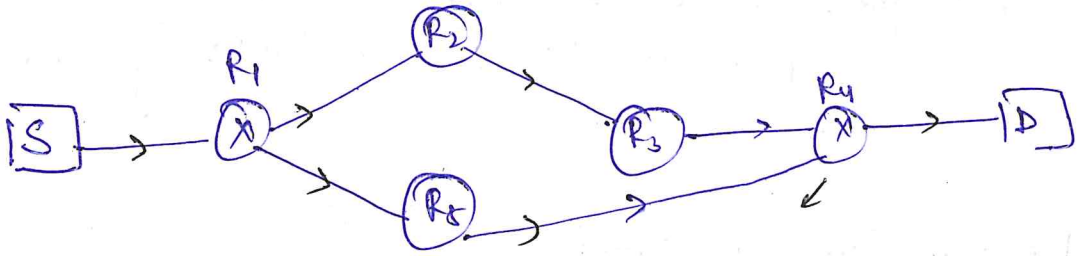
- Routing Algo
  - (Static) Non-adaptive
    - flooding
    - Random-walk.
  - Adaptive (Dynamic)
    - Distance-vector
    - Link state algo.

* Traffic changes over time and so congestion also changes. Topology also changes over time because of increment/decrement of router.

→ So there is change in graph therefore in Non-adaptive/static algorithms. we don't give worth to traffic and topology change.

* In Practical we use Adaptive/dynamic algorithms.

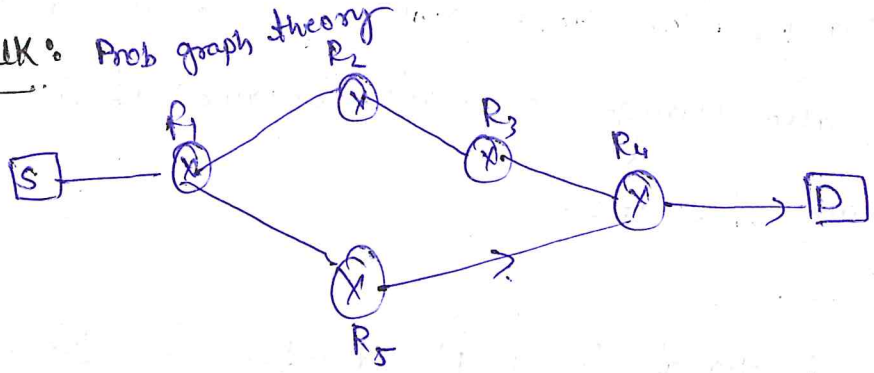
Flooding:



* In flooding whenever the router gets the packet it will flood to every other router in the path but except the point of origin.

* This will create congestion or looping.
* By flooding this is guarantee that if exists a single path from source to destination then the packet will reach.
* That makes flooding extremly reliable.
* Here we don't require any routing table.
* Atleast one packet will take the shortest path because we are taking every path.

Random walk: Prob graph theory



(1st) → Pick a router randomly [one technique] for sending the packet.

(2nd) technique: In this technique every router sends the packet to the least queued / busy router.

Distance-Vector-algo / routing

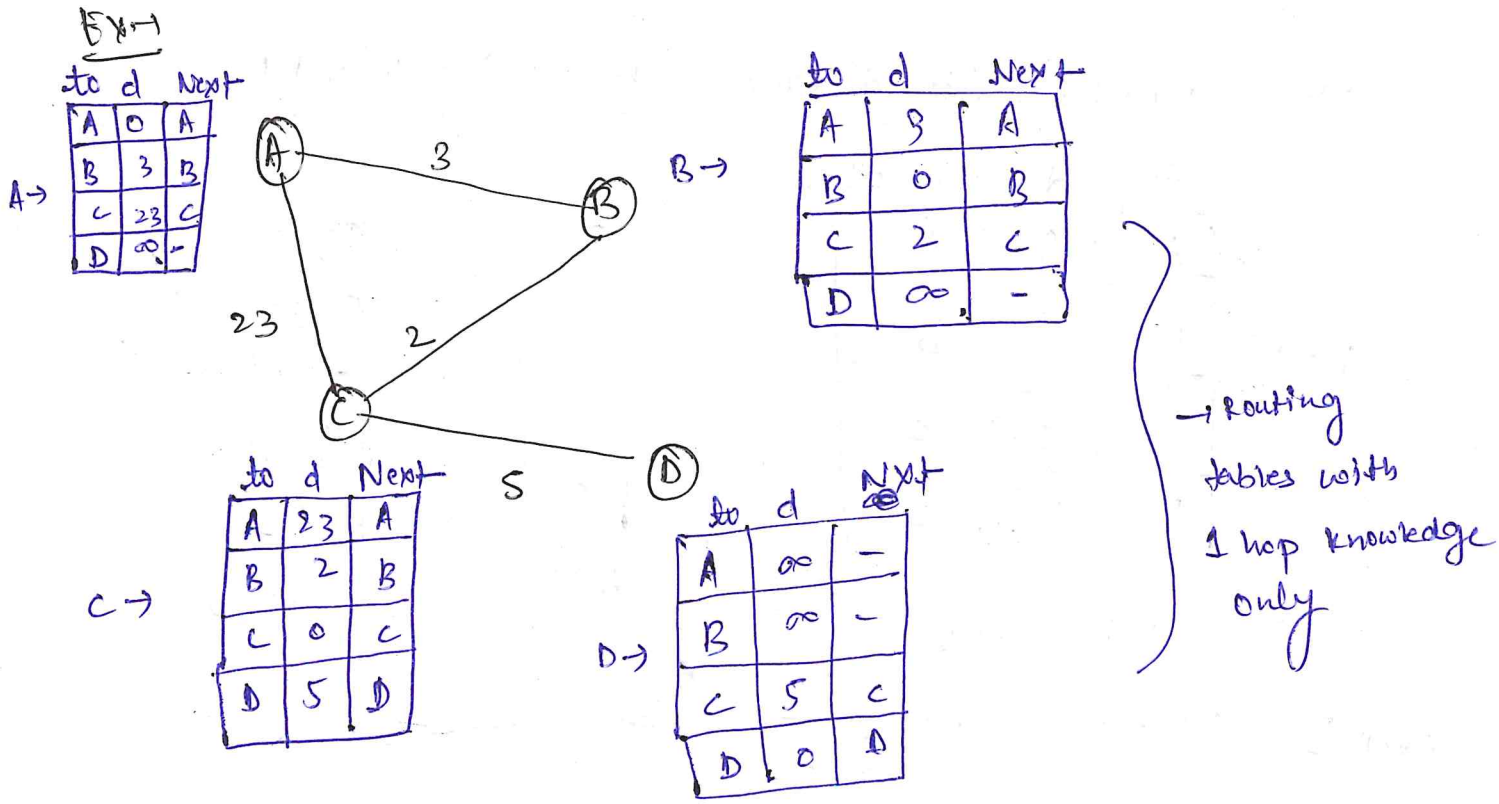
↳ Bellman-Ford-algo

↓

Subpaths of the shortest paths are shortest paths

If the shortest path u to v is given.
then the shortest path from u to w is also this.

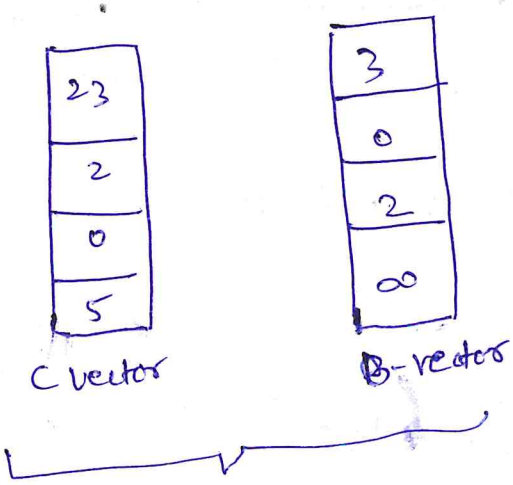
Ex→



At time $t = (1)$ → hop Every router A, B, C, D has the routing table and also knows its neighbour distances.

At time = (2) → hop we have to find the path length = 2

Now C, and B will send the distance vector to A



This information shared to A

Now, A updates its distance vector using B and C.

A →

to	d	next hop
A	0	A
B	3	B
C	5 ~~23~~	C
D	28	C

A new table

23
2
0
5

AC: 23

3
0
2
∞

AB: 3

AtoD = AtoC = 23 BtoC = 5 = ~~8~~ 23+5 = 28.

AtoC = AtoB = 3 BtoC = 2 = 5

B also get distance vector from A and C.

B →

to	d	Next	
A	3	A	✓
B	0	B	✓
C	2	C	✓
D	∞	f7	

B new table

0
3
23
∞

A distance vector

BA: 3

23
2
0
5

C-distance vector

BC: 2

B to D = B to A + A to D

= 3 + ∞

= ∞

B B to D = B to C + C to D

= 2 + 5 = 7

Similarly C will get distance vector from A and B

C new table

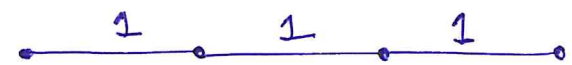
To	d	next
A	23	A
B	2	B
C	0	C
D	5	D

Similarly D will get distance vector from C

To	d	Next
A	28	C
B	7	C
C	5	C
D	0	D

* One iteration is completed.
* For distance vector we continue repeating this step because topology can change over period of time.

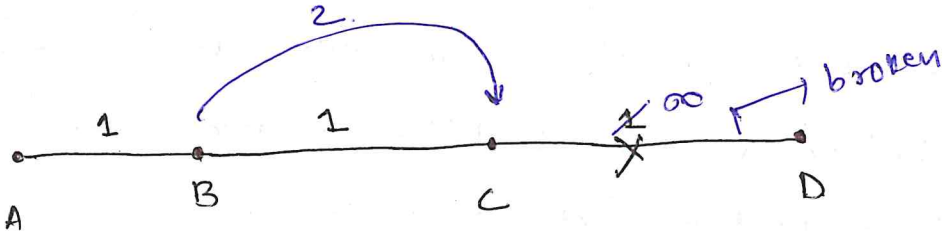
Count-to-∞ problem (topology changes)



[dist to D, Next]

(3,B) (2,C) (1,D) (0,D)

→ Here we are writing a distance to D from all routers.



3, B 2, C 1, D

Exch-1	3, B	2, C	3, B	⇒ C will update 3 because distance of D using B is 2
Exch→2	3, B	4, C	3, B	⇒ B update distance using C
Exch→3	5, B	4, C	5, B	⇒ A updates distance using B

These distances keep increasing upto ∞.

So, packet Stuck in loop ⇒ A → B → C → B → C ∞ - loop.

Solution for count to ∞ problem is split Horizon:

t	3, B	2, C	1, D
t+1	3, B	2, C	∞, -
t+2	3, B	∞, -	∞, -
t+3	∞, -	∞, -	∞, -

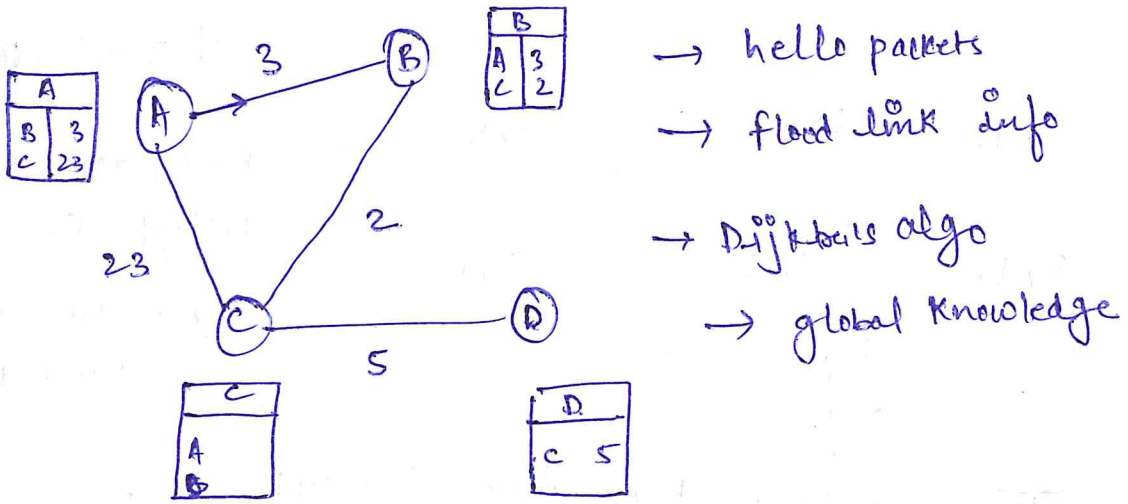
∞ (as 'B' depends on 'C' to goto D)

So, B will not transfer distance vector to C

→ for previous example of distance vector by using split horizon we only send the distance vector with ∞ to the next-hop node.

→ RIP uses distance vector routing

Link State Routing : (LSR)



→ hello packets

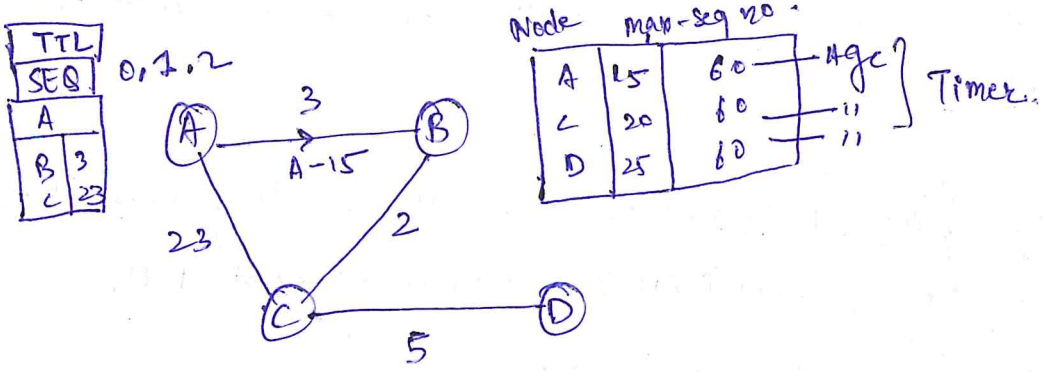
→ flood link info

→ Dijkstra's algo

→ global knowledge

* Every node construct a packet containing information about the adjacent nodes. These are called linked state packets.
* Every link state packet is flooded to every other node.
* Now on A → C, B, D packets will reach.
* Once we have a whole information of all nodes we can construct a graph and run dijstra's algorithm for shortest path.
* Every vertex has the knowledge of every other vertex.
* OSPF (Open Shortest Path first) most widely used algorithm for Link State.

Sequence number :



* Sequence number is help us to keep track how network is keep changing and evolving over time.

* We add sequence number because if any node receives the packet with older sequence no. that packet will be discarded.

* Every node has the list of maximum sequence number. If we receive the packet less than the max^m sequence number than that packet will be discarded. It also reduce the congestion by discarding the packet.

* We discard duplicate packet to reduce flooding.

* We have a TTL field in the packet. So passing through every router we can decrease the TTL value. This will help us avoid Transient looping (Temporary looping) (non-persistent)

* If a node re-started and sends a new fresh packet with sequence 0, the another node might think this is an old packet because of less sequence number.

The solution for this problem is Age field where. after some time like 60 sec age field will expire. and maxm sequence number entry will be remove from node.

Que: Consider the following statements about the routing protocols, Routing Information protocol (RIP) and Open Shortest Path first (OSPF) in an IPV4 Network.

I. RIP uses distance vector routing

II. RIP packets are sent using UDP

III. OSPF packets are sent using TCP.

IV. OSPF operation is based on link-state routing.

Which of the following above are CORRECT?

(A) I and IV only

(B) I, II, III only

(C) I, II, IV only

(D) II, III, IV only.

Ans → I → correct (we already discussed in video)

II → RIP used UDP (remember that)

III → X

IV → correct (we already mentioned in video)

Que: Suppose that in an IP-over Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following techniques can be used for this?

(a) X sends an ARP request packet with broadcast IP address in local subnet.

it sends with broadcast MAC address

(b) X sends an ARP request packet to the local gateways MAC address with then finds the MAC address of Y and sends to X

(c) X sends an ARP request packet with broadcast MAC address in its local subnet. [Already discussed in video]

(d) X sends an ARP request packet to the local gateway's IP address and then find the MAC address of Y and sends to X?

IPV6 addressing:-

Initially 32-bit Ip address 2^{32} addresses are possible.

→ IPV6 address = 128 bit

→ 2001 : 0db8 : 0000 : 0000 : 8a2e : 0370 : 7334.

16-bit 16-bit

Each digit is in hexadecimal form.

Address length = 16×8 = 128-bit.

→ 2001 : db8 : : 8a2e : 370 : 7334 {compressed format}

You can also remove zero prefix

If you have this double colon representation this means in between everything is zero

Note → You can use double colon only one time in the address.

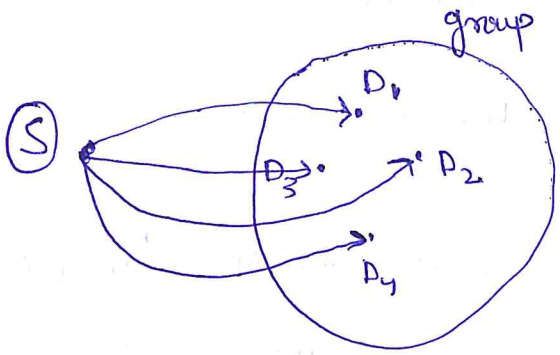
→ IPV4 and IPV6 are both used today.

UNICASTING:

Unicasting means single node to single node.

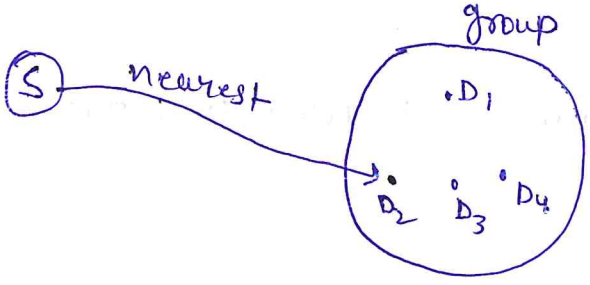
MULTCASTING:

Multicasting means if you want to send from single node to group of nodes.



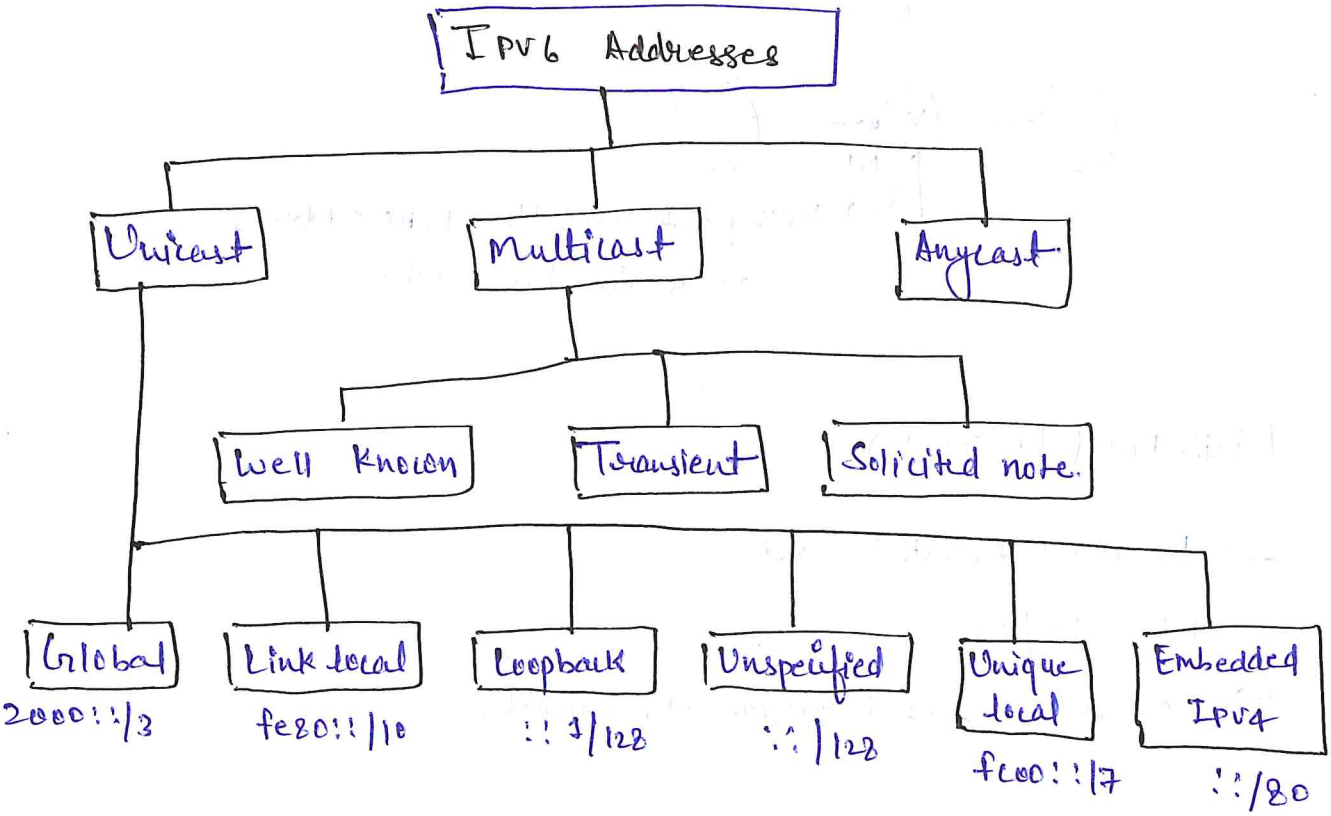
{ Broadcasting is a special case of Multicasting }

Anycasting:



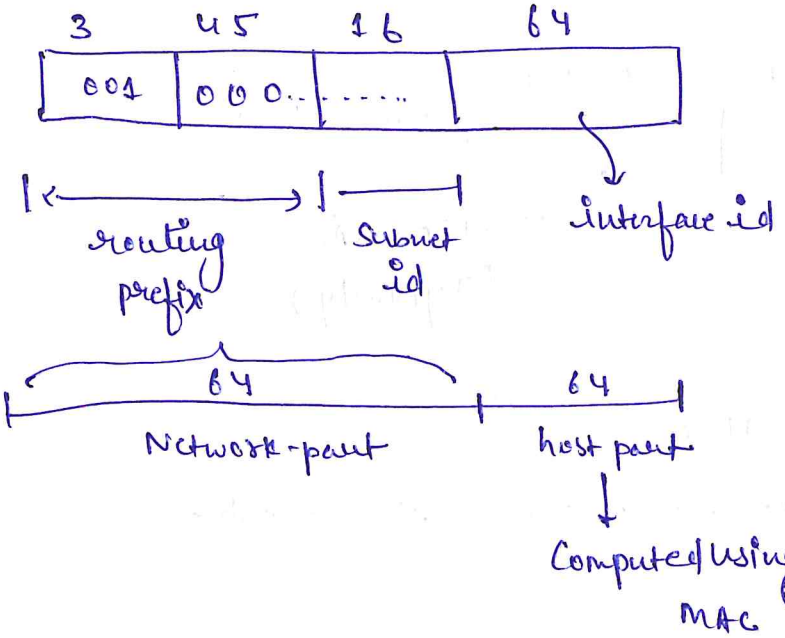
In Anycasting the objective is to send the packet from source to one of the node of the group.

* It is typically sent to the nearest host.

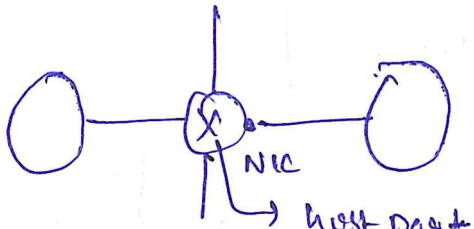


Global-unicast → Global unicast is just like public Ip addresses in IPV6.

Node can be host / router.



* In Global unicast address first three bits are 001 fixed.
* It is globally routable any router gets this will successfully routed to destinations.



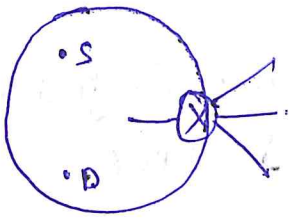
→ host part uses the MAC address as an interface id.

Link-Local (Unicast)

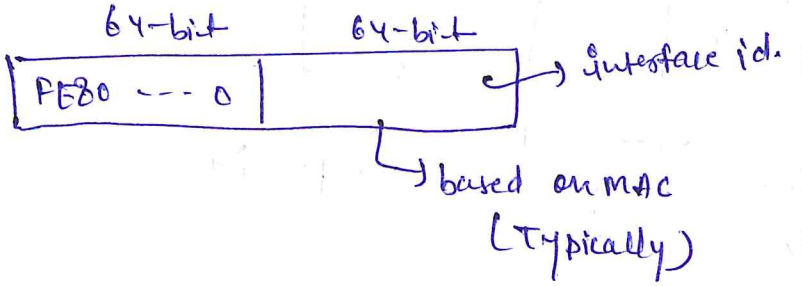
→ Local-network unicast

→ FE80 ::/64

→ Can't be routed outside the network.



If source and destination are in same network we use link local address.



* First bits are = FE80
* This packet will not route outside the network.

Loopback / Local host

→ ::1/128

* Loopback Ip address for testing / debugging.

Unspecified → ::/128 {all zeros}

→ not assigned to any interface

This is an unspecified Ip address

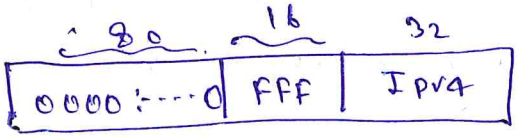
Unique Local → FC00::/7

→ Private Ip address just like IPV4

Embedded IPV4: → Transition period

IPV4
IPV6

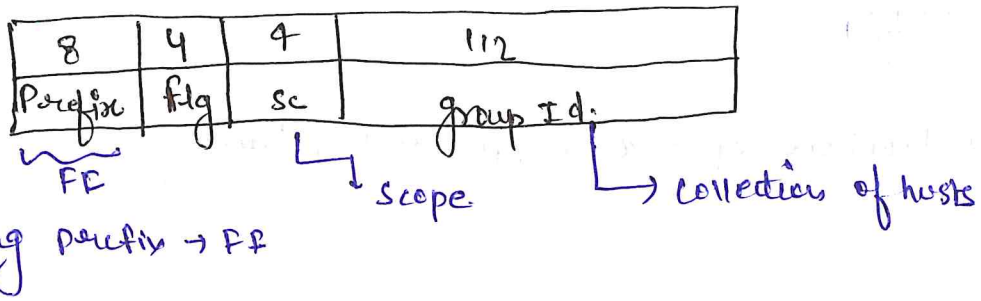
for converting IPV4 into the IPV6 address



→ 80 bits are zeroes, 16 bits are ffff, 32-bit IPV4 address

to convert IPV4 to IPV6.

Multicast:



Starting prefix → FF

* Multicasting is used for sending single node to group of nodes.

* Group Id is the collection of hosts, routers map group Id to hosts.

flag: 0000 → Permanent grouping

* Permant grouping can be used for fixed hosts like group of routers, gateways which are used by country.

flag: 0001 : Temporary / Transient

* We can create a temporary grouping for a meeting or company for some time.

Scope:

0000: reserved

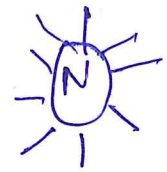
0001: Node local

0010: Link local

0101: Site local

1110: Global

1111: reserved.



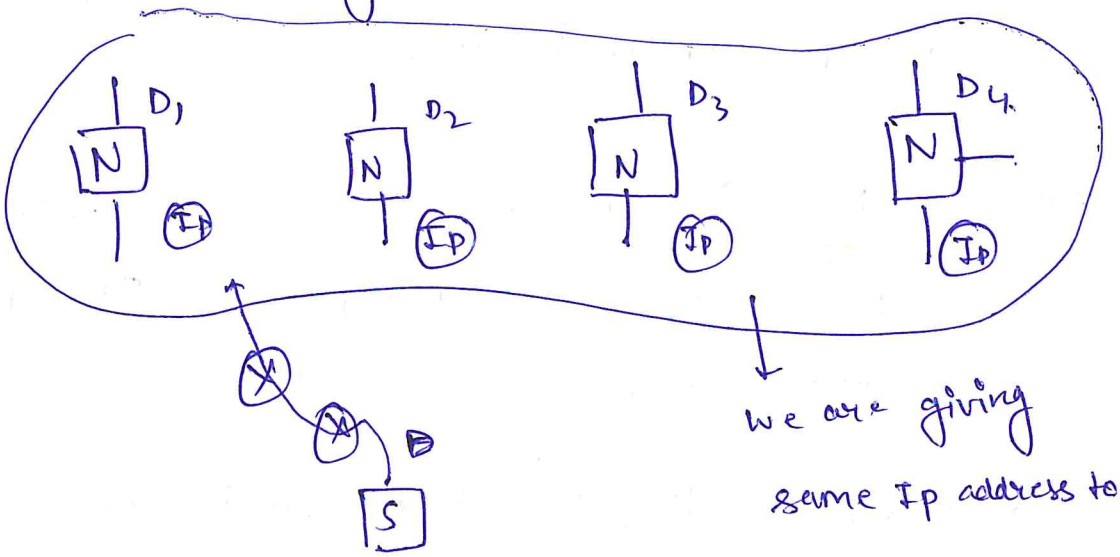
→ All the interface are local to the network

→ There are on the same geographical location

Anycasting:

→ No specific IPV6 range.

→ All Unicast address can be used for multicasting

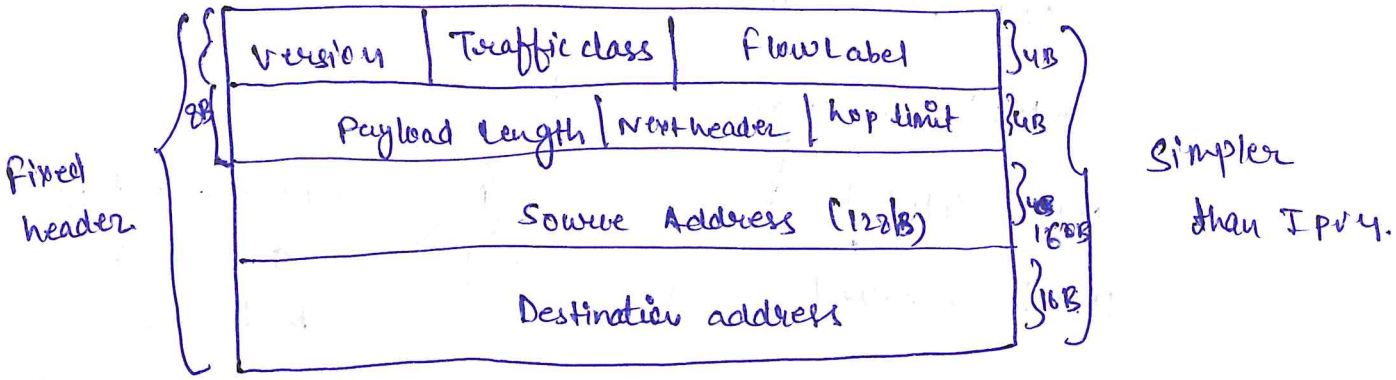


Anycasting

In IPV6 multiple interface has same Ip address.

* Source can send any of the interface using that Ip.

IPv-6 header :-



Extension - header - 1

Extension header - 2.

⋮

Data

* In IPV6 header there is fixed header followed by multiple extention header.

Version :- By seeing the version bits we can easily recognize whether it is an IPv4 packet or IPV_6 packet.

0110 → IPV_6.

* 40 Bytes is the size of the fixed header.

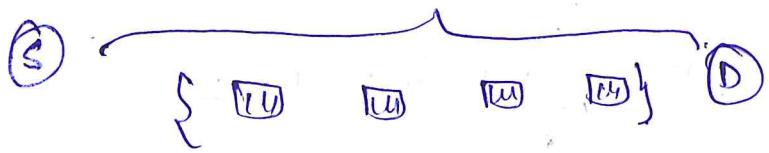
Traffic class:

- Same as (DSCP + ECN) fields (TOS) in IPV4.
- Pariority, congestion-notification

Flow-label: Its a 20-bit field.

The default value is 0.

* From source to destination there is sequence of packets that we are sending from saurce to destination.
  - TCP packets of same flow/stream.
* Groups of packets are part of the same flow.



* routers will give the speial servire to these packets.

Payload length: — Extention-headers + Data (from layers above)
(16 bit)

If length of payload ≥ 65535

* Then, Set the payload length to zero and use hop by hop extention header using Jumbo payload option.

Next header → TCP | UDP | ICMPv6 | extention-header type.

It stores the protocol id of the data header.

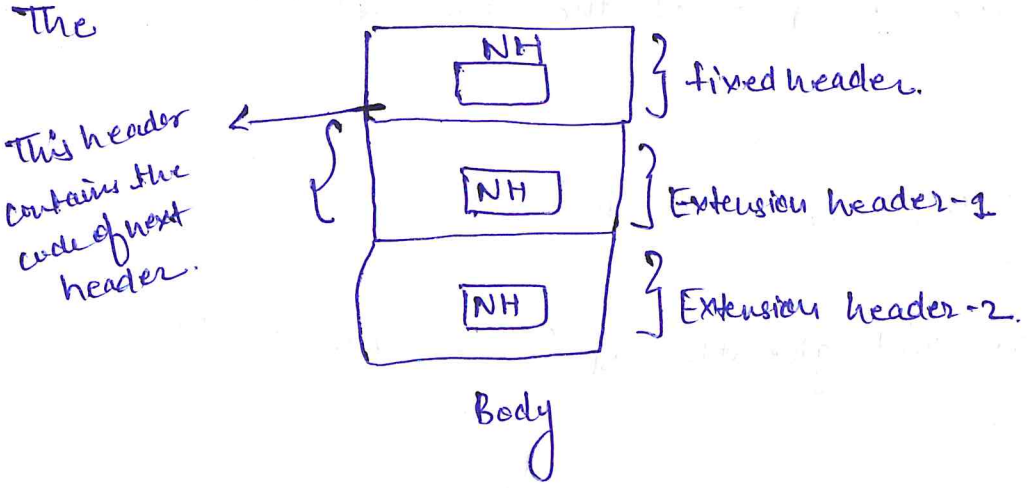
* If data contains TCP packet then next header contains id of the TCP.

Hop limit → Exactly like TTL in IPv4

Extention header: Initially we have a fixed header and afterwards will have extension header :1.

The extension header contains the extra functioning that you want in packet.

The



The order of extension header is

Recommended order

order	Header Type	Next Header code
1.	Basic IPv6 Header	-
2.	Hop by hop options	0
3.	Destination options	60

} like that we follow heads in extension header

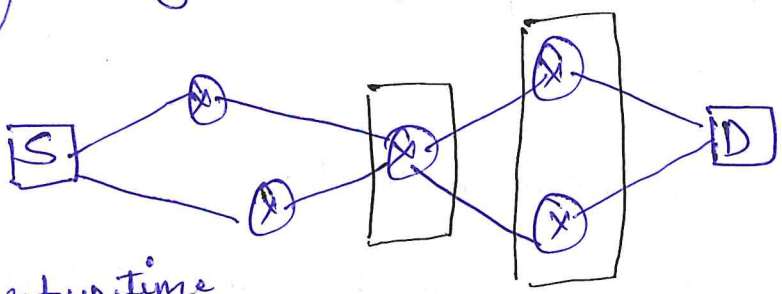
Switching:

- Circuit Switching (Telephone)
- Packet switching
  - Datagram (IP)
  - Virtual circuits (ATM) Async. Transfer mode
- Message Switching (Military application)

Circuit Switching:-

→ Telephone - exchange (1960s)

VOLTE



→ connection set up time

↳ Here we call exchange to establish the connection upto the destination.

→ No headers, no store and forward

→ data arrives in order

→ Message size: m.

→ $\left[\text{Setup time} + \underbrace{\frac{m}{B}}_{T_t} + \underbrace{\frac{\text{dist}}{\text{vel}}}_{T_p} + \text{Teardown time} \right]$

→ Once a dedicated connection is set up, we can share the informations from source to destination.

→ data arrives in order in the circuit switch model.

→ In datagram model, packet need not arrive in order.

[

Packet Switching :—

→ Datagrams (IPV4) {routers}

→ Virtual circuits (ATM)

* Only 1st packet has header
* Same path by all packets (Resource reservation)

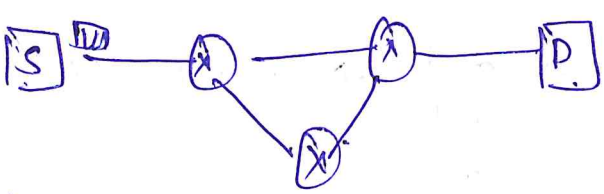
→ INORDER.

→ Reliable and connection-oriented.

* In packet switching we breack the data and form chunk of packet for sending.

* Datagram is the most popular in packet switching.

* Virtual circuits (Asynchronous transfer mode)

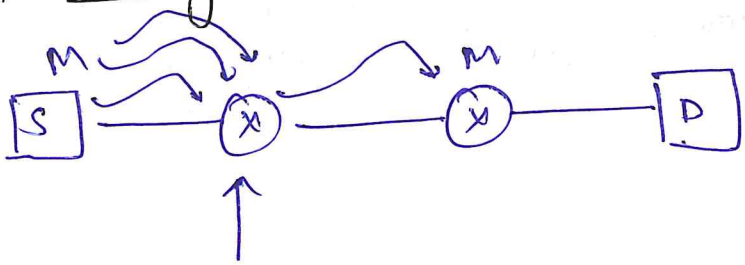


↳ It is using functioning of circuit somewhat

* The first packet in the intermediate routers tells us all the intermediate routers to reserve some space for coming packets.

* Virtual circuits are reliable and connection oriented.

Message Switching :-



{ Complete Message
Store and forward.

Message completely stored to the next hop and then it is forwarded.

→ It is used in Military applications and satellite communication.